

Supplement to “Event-aligned conditional quantile trajectories of a biomarker before disease onset under left truncation and competing death”

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This supplement collects the proofs (Section S1), the estimation algorithms and implementation details (Section S2), the full set of simulation and audit/sensitivity studies summarized in Section 5 of the main paper (Section S3), and additional details of the application (Section S4). Equations, tables, and figures are numbered S1, S2, $\dots$; references of the form “Theorem 1” point to the main paper.

S1 Proofs

This section reproduces the asymptotic theory: the transport/identification arguments, the implicit density-weighted orthogonalization, the censoring orthogonality ($\chi^C = 0$), the sharper-remainder bound, the high-level cross-fitted one-step asymptotic normality and sandwich consistency, and the clustered, weighted, growing-sieve Gram/Knight/Bahadur representation that verifies the high-level conditions for the concrete sieve. The numbering of internal results is self-contained within this section.

S1.A Identification, orthogonality, one-step normality, sandwich

Status of this document. This supplement is written for a *generic* observed-event weight $\omega(T, A, \mathbf{X})$, of which the default complete-event ($\omega \equiv 1$) and the IPCW ($\omega = 1/G_C$) members are special cases. §S1.0.10–§S1.0.2 are *complete and self-contained*: the observed-data transport (Lemma 1), generic-weight identification (Proposition 1), the hierarchical Neyman orthogonality to (m, g, f) for every member and the additional G_C direction for the IPCW member (Lemmas 2–5), and the bread symmetrization (Corollary 1). §S1.0.3 *proves* the sharper population-drift remainder (Lemma 6) carrying the generic weight perturbation δ_ω , and Proposition 3 gives the product form of the projected sieve bias. §S1.0.4 states and proves the *high-level* cross-fitted one-step theorem (Theorem 2) under explicit nuisance-rate conditions, with the cross-fitted L_2 score-reduction lemma (Lemma 7) and the sandwich-consistency corollary (Corollary 2). The verification that a concrete B-spline / quantile-spacing sieve estimator *attains* the rate conditions assumed here is the separate task of supplement II; we flag the boundary between what is proved here and what is deferred there.

S1.0.1 Setup: generic observed-event weight

For a subject with $\Delta^T = 1$ (onset observed) write $H = (A, \mathbf{X})$, $W = T - A$, $L_\ell = \log Y(U_\ell)$, $\varepsilon_\ell(\boldsymbol{\beta}, m) = L_\ell - m(U_\ell, T) - \mathbf{X}^\top \boldsymbol{\beta}$, $\psi_\tau(r) = \tau - \mathbf{1}\{r \leq 0\}$, and $f_\ell(0) = f_{\varepsilon|U, T, \mathbf{X}, D \geq T}(0 | U_\ell, T, \mathbf{X})$ for the conditional residual density at its τ -quantile. Let $\omega(T, A, \mathbf{X})$ be a *bounded, strictly positive* observed-event weight,

$$0 < c_\omega \leq \omega(T, A, \mathbf{X}) \leq C_\omega < \infty, \quad a_\omega = \Delta^T \omega(T, A, \mathbf{X}). \quad (\text{S1})$$

The default *complete-event* (CC) member is $\omega \equiv 1$; the *IPCW* member is $\omega = 1/G_C(W | H)$, finite under censoring positivity $G_{C,0} \geq c_G > 0$.

Convention. Population identities (the occupation measures, M , orthogonality, drift) are stated under the *target-population* law P_0 , with all selection (left-truncation retention S , onset-observation Δ^T) carried *inside* indicators and weights, never by conditioning on $\{S = 1\}$ (see “Two laws” below); the retained-sample law $P_S = P_0(\cdot | S = 1)$ enters only when counting i.i.d. subjects for the

empirical process. The selection $T \leq D$ is carried by Δ^T inside a_ω , so $\mathbf{1}\{T \leq D\}$ appears explicitly only *after* the censoring identity (Assumption 2 of the main text), $\mathbb{E}_0[\Delta^T | Y, T, D, A, \mathbf{X}, N^{V,0}] = \mathbf{1}\{T \leq D\} G_{C,0}(W | H)$; we never double-condition. Nuisances are $\eta = (m, g, f, G_C)$. The moment functional, valued in $\mathbb{R}^p$, is

$$M(\boldsymbol{\beta}, m, g; \omega) = \mathbb{E}\left[a_\omega \sum_{\ell} \{\mathbf{X} - g(U_\ell, T)\} \psi_\tau(\varepsilon_\ell(\boldsymbol{\beta}, m))\right], \quad (\text{S2})$$

which does *not* depend on f . Define the population *design weight* that arises after the censoring identity,

$$r_\omega = G_{C,0}(W | H) \omega(T, A, \mathbf{X}), \quad (\text{S3})$$

so $r_{CC} = G_{C,0}$ and $r_{IPCW} \equiv 1$. Define the scalar and vector *visit-occupation measures* on $\{0 < u < t\}$,

$$\nu_{\tau,\omega}(B) = \mathbb{E}\left[\mathbf{1}\{T \leq D\} r_\omega \sum_{\ell} f_\ell(0) \mathbf{1}\{(U_\ell, T) \in B\}\right], \quad (\text{S4})$$

$$\mu_{\tau,\omega}(B) = \mathbb{E}\left[\mathbf{1}\{T \leq D\} r_\omega \sum_{\ell} f_\ell(0) \mathbf{X} \mathbf{1}\{(U_\ell, T) \in B\}\right] \in \mathbb{R}^p, \quad (\text{S5})$$

and, assuming $\mu_{\tau,\omega} \ll \nu_{\tau,\omega}$ componentwise, the *density-weighted projection* $g_{\tau,\omega} = d\mu_{\tau,\omega}/d\nu_{\tau,\omega}$ ($\nu_{\tau,\omega}$ -a.e.). The information matrix and the (member-specific) influence function are

$$S_{\tau,\omega} = \mathbb{E}\left[\mathbf{1}\{T \leq D\} r_\omega \sum_{\ell} f_\ell(0) \{\mathbf{X} - g_{\tau,\omega}(U_\ell, T)\}^{\otimes 2}\right], \quad \phi_{i,\tau,\omega}^0 = a_\omega \sum_{\ell} \{\mathbf{X} - g_{\tau,\omega}(U_\ell, T)\} \psi_\tau(\varepsilon_{0,\ell}), \quad (\text{S6})$$

$\varepsilon_{0,\ell} = L_\ell - m_0(U_\ell, T) - \mathbf{X}^\top \boldsymbol{\beta}_{\tau,0}$. The weight ω enters $g_{\tau,\omega}$, $S_{\tau,\omega}$, and ϕ^0 , so each member has its *own* projection, bread, and efficiency; the estimand $\boldsymbol{\beta}_{\tau,0}$ is common (Proposition 1).

Estimator (fold-specific cross-fitting). K -fold subject splitting; on the complement of fold k fit *all* of $\widehat{m}^{(-k)}, \widehat{f}^{(-k)}, \widehat{g}^{(-k)}, \widehat{\omega}^{(-k)}$ and an initial $\widehat{\boldsymbol{\beta}}_{\text{init}}^{(-k)}$, then evaluate on fold k . With $\widehat{\phi}_i^{(-k)}(\boldsymbol{\beta}) = \widehat{a}_i^{(-k)} \sum_{\ell} \{\mathbf{X}_i - \widehat{g}^{(-k)}(U_{i\ell}, T_i)\} \psi_\tau(\varepsilon_{i\ell}(\boldsymbol{\beta}, \widehat{m}^{(-k)}))$, $\widehat{a}_i^{(-k)} = \Delta_i^T \widehat{\omega}^{(-k)}$, and $\widehat{S}_k = \mathbb{P}_{n,k}[\widehat{a}^{(-k)} \sum_{\ell} \widehat{f}_\ell^{(-k)}(0) \{\mathbf{X} - \widehat{g}^{(-k)}\}^{\otimes 2}]$, the cross-fitted one-step is the *sample-size-weighted* fold average

$$\widetilde{\boldsymbol{\beta}}_{\tau,\omega} = \sum_{k=1}^K \frac{n_k}{n} \left[\widehat{\boldsymbol{\beta}}_{\text{init}}^{(-k)} + \widehat{S}_k^{-1} \mathbb{P}_{n,k} \widehat{\phi}^{(-k)}(\widehat{\boldsymbol{\beta}}_{\text{init}}^{(-k)}) \right], \quad K \text{ fixed, } \min_k n_k/n \geq c_K > 0, \quad (\text{S7})$$

so $\sum_k (n_k/n) \mathbb{P}_{n,k} \phi = \mathbb{P}_n \phi$ exactly. Every nuisance and the initial estimator entering fold k are trained off fold k .

S1.0.2 Observed-data transport and identification

The structural model (main text §2) specifies the conditional-quantile restriction on the *full-data*, *target-population* residual,

$$\mathbb{E}\{\psi_\tau(\varepsilon^0(u)) | T^0 = t, \mathbf{X}^0, D^0 \geq t\} = 0, \quad (\text{S8})$$

i.e. $m_{\tau,0}(u, t) + \mathbf{X}^\top \boldsymbol{\beta}_{\tau,0}$ is the conditional τ -quantile of $L^0(u)$ given $(T^0 = t, \mathbf{X}^0, D^0 \geq t)$. Every orthogonality and drift result below uses instead the *observed-visit* moment

$$\mathbb{E}\{\psi_\tau(\varepsilon_\ell) | U_\ell, T, \mathbf{X}, D \geq T, \Delta^T = 1\} = 0, \quad (\text{S9})$$

which is not assumed but must be *transported* from (S8).

Two laws. Let P_0 be the full target-population law and $N^{V,0}$ the pre-onset visit-counting process. We state all population identities under P_0 , carrying retention/onset selection *inside* indicators and weights rather than by conditioning on $\{S = 1\}$ or on the measure-zero event $\{U = u\}$; the retained-sample law $P_S = P_0(\cdot | S = 1)$ is used only when counting i.i.d. subjects in the empirical process. Because U is continuous, $P_0(U = u) = 0$, so the restriction must be stated in *integrated* (occupation/Campbell) form, matching the definition of $\nu_{\tau,\omega}$ in (S4).

Lemma 1 (Transport via the visit-occupation measure). *Assume (i) $A \perp \{L^0(\cdot), T, D\} | \mathbf{X}$; (ii) $N^{V,0} \perp \{L^0(\cdot), T, D\} | A, \mathbf{X}$; (iii) $C \perp \{L^0(\cdot), T, D, N^{V,0}\} | A, \mathbf{X}$; (iv) positivity on the estimable support. Then for every bounded measurable $a(u, t, \mathbf{X})$ and the design weight r_ω of (S3),*

$$\mathbb{E}_0 \left[\mathbf{1}\{T \leq D\} r_\omega \int_0^T a(u, T, \mathbf{X}) \psi_\tau \{\varepsilon^0(u)\} dN^{V,0}(u) \right] = 0, \quad (\text{S10})$$

where $\varepsilon^0(u) = L^0(u) - m_{\tau,0}(u, T) - \mathbf{X}^\top \beta_{\tau,0}$. Equivalently, disintegrating against the visit-occupation measure $\nu_{\tau,\omega}$ of (S4), $\mathbb{E}_{\nu_{\tau,\omega}}[\psi_\tau(\varepsilon) | U, T, \mathbf{X}] = 0$ for $\nu_{\tau,\omega}$ -a.e. $(u, t, \mathbf{X})$. This is the integrated observed-visit restriction; no conditioning on $\{U = u\}$ is used.

Proof. The observed weight is $a_\omega = \Delta^T \omega$ with $\omega = \omega(T, A, \mathbf{X})$. Condition the integrand of the (pre-censoring) functional on the full vector $(A, \mathbf{X}, T, D, N^{V,0}, L^0(\cdot))$, against which the visit integral is measurable; by independent censoring (iii), $\mathbb{E}_0[\Delta^T | A, \mathbf{X}, T, D, N^{V,0}, L^0(\cdot)] = \mathbf{1}\{T \leq D\} G_{C,0}(W | H)$, and since ω is $(T, A, \mathbf{X})$ -measurable this turns a_ω into $\mathbf{1}\{T \leq D\} r_\omega$, giving the integrand of (S10). Now condition on $(A, \mathbf{X}, T, D)$. By (ii) the visit measure $N^{V,0}$ is conditionally independent of $L^0(\cdot)$ given $(A, \mathbf{X})$, so by the Campbell/Mecke formula the inner integral has conditional mean

$$\int_0^T a(u, T, \mathbf{X}) \mathbb{E}_0[\psi_\tau \{\varepsilon^0(u)\} | A, \mathbf{X}, T, D] \mathbb{E}_0[dN^{V,0}(u) | A, \mathbf{X}],$$

and by (i) entry A does not select the marker given $(T, D, \mathbf{X})$, so $\mathbb{E}_0[\psi_\tau \{\varepsilon^0(u)\} | A, \mathbf{X}, T, D] = \mathbb{E}_0[\psi_\tau \{\varepsilon^0(u)\} | T, \mathbf{X}, D \geq T]$ on $\{T \leq D\}$, which is 0 by the structural restriction (S8). Hence the inner integral has conditional mean 0, and (S10) follows by taking outer expectation. The disintegration statement is the Radon–Nikodym factorization of the bounded linear functional $a \mapsto$ (S10)’s integrand against $\nu_{\tau,\omega}$ (well defined by positivity (iv)); since it vanishes for every bounded a in a determining class, $\mathbb{E}_{\nu_{\tau,\omega}}[\psi_\tau(\varepsilon) | U, T, \mathbf{X}] = 0$ $\nu_{\tau,\omega}$ -a.e. $\square$

Throughout, “the observed-visit moment” (S9) is shorthand for the rigorous occupation-measure identity of Lemma 1; every later use is as an integral against $\nu_{\tau,\omega}$ or $\mu_{\tau,\omega}$, never a pointwise conditioning.

Proposition 1 (Generic-weight moment invariance). *Under Lemma 1, for any bounded measurable g and any weight ω satisfying (S1), $M(\beta_{\tau,0}, m_0, g; \omega) = 0$.*

Proof. Take $a(u, t, \mathbf{X}) = \{\mathbf{X} - g(u, t)\}$ (componentwise) in Lemma 1’s identity (S10): the visit integral of $\{\mathbf{X} - g\} \psi_\tau(\varepsilon^0)$ against $\mathbf{1}\{T \leq D\} r_\omega dN^{V,0}$ has expectation 0, which is exactly $M(\beta_{\tau,0}, m_0, g; \omega)$ written in occupation form. Hence $M = 0$ regardless of g and of the positive weight ω . This is *moment unbiasedness, not identification*; the p -vector M cannot by itself pin down the infinite-dimensional m . $\square$

Proposition 2 (Identification of $(m_{\tau,0}, \beta_{\tau,0})$). *Let $Q_\omega(m, \beta) = \mathbb{E}[\Delta^T \omega \sum_\ell \rho_\tau \{L_\ell - m(U_\ell, T) - \mathbf{X}^\top \beta\}]$ be the population weighted check-loss risk. Assume, in addition to Lemma 1: the conditional*

residual density is strictly positive in a neighborhood of 0 on the estimable support; $\mathbf{X}$ contains no intercept and no deterministic function of (u, t) ; and the covariate Gram $\mathbb{E}_{\nu_{\tau, \omega}}[\{\mathbf{X} - \mathbb{E}_{\nu}[\mathbf{X} | U, T]\}^{\otimes 2}]$ is nonsingular. Then $(m_{\tau, 0}, \beta_{\tau, 0})$ is the unique minimizer of Q_{ω} over the estimable support.

Proof. By Knight's identity, with $d_{\ell} = m(U_{\ell}, T) - m_0(U_{\ell}, T) + \mathbf{X}^{\top}(\beta - \beta_{\tau, 0})$,

$$Q_{\omega}(m, \beta) - Q_{\omega}(m_0, \beta_{\tau, 0}) = \mathbb{E} \left[\Delta^T \omega \sum_{\ell} \left(-\psi_{\tau}(\varepsilon_{0, \ell}) d_{\ell} + \int_0^{d_{\ell}} \{F_{\ell}(s | Z) - F_{\ell}(0 | Z)\} ds \right) \right].$$

The first term has expectation 0 by Lemma 1 (moment invariance applied with $a = d$). The integral is nonnegative because $F_{\ell}(\cdot | Z)$ is nondecreasing and $\varepsilon_{0, \ell}$ has its τ -quantile at 0 (so $F_{\ell}(s) - F_{\ell}(0)$ has the sign of s); hence $Q_{\omega}(m, \beta) \geq Q_{\omega}(m_0, \beta_{\tau, 0})$. Equality forces the integral to vanish, and by strict positivity of the density near 0 this requires $d_{\ell} = 0$, i.e. $m(U_{\ell}, T) + \mathbf{X}^{\top} \beta = m_0(U_{\ell}, T) + \mathbf{X}^{\top} \beta_{\tau, 0}$, $\nu_{\tau, \omega}$ -a.e. Since $\mathbf{X}$ carries no intercept or (u, t) -function and the covariate Gram (after projecting out the (U, T) -measurable part) is nonsingular, the (u, t) -part and the $\mathbf{X}$ -part separate: $m = m_0$ $\nu_{\tau, \omega}$ -a.e. and $\beta = \beta_{\tau, 0}$. $\square$

Hence every member estimates the same $(m_{\tau, 0}, \beta_{\tau, 0})$; CC ($\omega \equiv 1$) and IPCW ($\omega = 1/G_C$) differ only in projection, information, and efficiency.

$D_{\eta}M[\cdot]$ and $D_{\beta}M$ below are Gateaux derivatives at $(\beta_{\tau, 0}, \eta_0)$; $D_{\beta}M = -S_{\tau, \omega}$ (Corollary 1).

Lemma 2 (Orthogonality to the surface m). *For bounded measurable h , $D_m M[h] = -\int h (d\mu_{\tau, \omega} - g d\nu_{\tau, \omega})$. If $g = g_{\tau, \omega} = d\mu_{\tau, \omega}/d\nu_{\tau, \omega}$ then $D_m M[h] = 0$ for all such h ; conversely $D_m M[h] = 0$ over a class dense in $L_2(\nu_{\tau, \omega})$ with $\mu_{\tau, \omega} \ll \nu_{\tau, \omega}$ forces $g = g_{\tau, \omega}$ $\nu_{\tau, \omega}$ -a.e.*

Proof. Perturb $m_0 \mapsto m_0 + sh$. By the conditional-c.d.f. expansion at the τ -quantile (using Lemma 1 so the relevant law is the observed-visit one), $\partial_s \mathbb{E}[\psi_{\tau}(\varepsilon_{\ell}(\beta_{\tau, 0}, m_0 + sh)) | U_{\ell}, T, \mathbf{X}, D \geq T] |_{s=0} = -f_{\ell}(0)h(U_{\ell}, T)$. Hence, after the censoring identity, $D_m M[h] = -\mathbb{E}[\mathbf{1}\{T \leq D\} r_{\omega} \sum_{\ell} f_{\ell}(0) \{\mathbf{X} - g(U_{\ell}, T)\} h(U_{\ell}, T)] = -\int h (d\mu_{\tau, \omega} - g d\nu_{\tau, \omega})$. The forward and converse claims are the Radon-Nikodym characterization on the stated direction class. $\square$

Lemma 3 (Orthogonality to the centering g). *For bounded $\mathbb{R}^p$ -valued k , $D_g M[k] = -\mathbb{E}[a_{\omega} \sum_{\ell} k(U_{\ell}, T) \psi_{\tau}(\varepsilon_{0, \ell})] = -\mathbb{E}[\mathbf{1}\{T \leq D\} r_{\omega} \sum_{\ell} k(U_{\ell}, T) \mathbb{E}\{\psi_{\tau}(\varepsilon_{0, \ell}) | U_{\ell}, T, \mathbf{X}, D \geq T, \Delta^T = 1\}] = 0$, for any g , by Lemma 1.*

Lemma 4 (Orthogonality to the density f). *$D_f M \equiv 0$, as f does not appear in (S2). The true $g_{\tau, \omega} = g_{\tau, \omega}(f)$ depends on f , but the total derivative along $f_s = f_0 + sh_f$ (with $g_s = g_{\tau, \omega}(f_s)$) is $D_f M[h_f] + D_g M[\dot{g}_f] = 0 + 0 = 0$ by Lemma 3. Hence $\hat{f}$ enters no first-order term; it acts only through $\hat{g}$ (second order, §S1.0.3) and the bread $\hat{S}$ (consistency only).*

Lemma 5 (Orthogonality to censoring G_C — IPCW member only). *For the IPCW member $\omega = 1/G_C$, $D_{G_C} M[h] = -\mathbb{E}[\mathbf{1}\{T \leq D\} \xi_{\tau} h(W | H)/G_{C, 0}(W | H)] = 0$, where $\xi_{\tau} = \sum_{\ell} \{\mathbf{X} - g_{\tau, \omega}\} \psi_{\tau}(\varepsilon_{0, \ell})$, by differentiating $1/G_C$, conditioning on the full data, and $\mathbb{E}[\xi_{\tau} | T, A, \mathbf{X}, D \geq T, \{U_{\ell}\}] = 0$ (Lemma 1). The CC member does not estimate G_C , so this direction is vacuous there: CC is three-way orthogonal (m, g, f) , IPCW four-way.*

Corollary 1 (Bread symmetrization). $\mathbb{E}[\mathbf{1}\{T \leq D\} r_{\omega} \sum_{\ell} f_{\ell}(0) \{\mathbf{X} - g_{\tau, \omega}\} g_{\tau, \omega}^{\top}] = 0$, hence $S_{\tau, \omega} = \mathbb{E}[\mathbf{1}\{T \leq D\} r_{\omega} \sum_{\ell} f_{\ell}(0) \{\mathbf{X} - g_{\tau, \omega}\} \mathbf{X}^{\top}] = \mathbb{E}[\mathbf{1}\{T \leq D\} r_{\omega} \sum_{\ell} f_{\ell}(0) \{\mathbf{X} - g_{\tau, \omega}\}^{\otimes 2}]$, and $D_{\beta} M = -S_{\tau, \omega}$.

Proof. $g_{\tau, \omega}$ is (U, T) -measurable, so $\mathbb{E}[\mathbf{1}\{T \leq D\} r_{\omega} \sum_{\ell} f_{\ell}(0) \{\mathbf{X} - g_{\tau, \omega}\} g_{\tau, \omega}^{\top}] = \int g_{\tau, \omega}^{\top} (d\mu_{\tau, \omega} - g_{\tau, \omega} d\nu_{\tau, \omega}) = 0$ by $g_{\tau, \omega} = d\mu_{\tau, \omega}/d\nu_{\tau, \omega}$. Subtracting from $\mathbb{E}[\cdots \{\mathbf{X} - g_{\tau, \omega}\} \mathbf{X}^{\top}]$ gives the two equal forms. $\square$

Theorem 1 (Hierarchical orthogonality). *At $(\beta_{\tau,0}, \eta_0)$ with $g = g_{\tau,\omega}$, $D_m M = D_g M = D_f M = 0$ for every member, and $D_{G_C} M = 0$ additionally for the IPCW member; and $D_\beta M = -S_{\tau,\omega}$. The candidate influence function is $S_{\tau,\omega}^{-1} \phi_{i,\tau,\omega}^0$. This is local orthogonality, not multiple robustness under union-model misspecification: an $O(1)$ -misspecified $m_\tau \neq m_0$ breaks identification of β_τ even with $g_{\tau,\omega}, f, G_C$ correct.*

S1.0.3 Sharper population drift and projected sieve bias

Write $\delta_m = m - m_0$, $\delta_g = g - g_{\tau,\omega}$, $\delta_\beta = \beta - \beta_{\tau,0}$, $\delta_\omega = \omega - \omega_0$, the residual shift $h_\ell = \delta_m(U_\ell, T) + \mathbf{X}^\top \delta_\beta$, and the $L_2(\nu_{\tau,\omega})$ sizes $r_m = \|\delta_m\|$, $r_g = \|\delta_g\|$, $r_\beta = \|\delta_\beta\|$ (Euclidean), $r_\omega = \|\delta_\omega\|_\infty$.

Assumption 1 (Standing boundedness for the remainder/score lemmas). $\|\mathbf{X}\| \leq C_X$, $\sup_{u,t} \|g_{\tau,\omega}(u, t)\| \leq C_g$, the weights satisfy (S1), and with probability $\rightarrow 1$ the nuisance estimates lie in a fixed bounded set ($\|\hat{g}^{(-k)}\|_\infty \leq C'_g$, $\|\hat{\omega}^{(-k)}\|_\infty \leq C_\omega$); the within-subject visit count obeys $\mathbb{E}[M_i^2] < \infty$ (clean version: $M_i \leq \bar{M}$ a.s., or $M_i(1 + \|\mathbf{X}_i\|)^2$ sub-exponential, used in Supplement II). Under these the cross-visit sums are controlled by $(\sum_{\ell \leq M_i} q_{i\ell})^2 \leq M_i \sum_{\ell \leq M_i} q_{i\ell}^2$, and $L_2(\nu_{\tau,\omega})$ and the score-law $L_2(P)$ are equivalent norms (bounded densities/weights), so $L_2(\nu)$ rates transfer to $L_2(P)$.

A general finite-moment alternative replaces the L_2 bounds by L_4 : with $r_{d,4} = \{\mathbb{E}[\mathbf{1}\{T \leq D\} r_\omega \sum_\ell |d_\ell|^4]\}^{1/4}$ the bounds below hold with r_d^2 replaced by $r_{d,4}^2$ and an extra cluster-moment factor; we state the clean bounded version and flag the L_4 route.

Lemma 6 (Sharper population-drift bound, generic weight). *Assume the conditional-quantile model, Lemma 1, weight positivity (S1), the boundedness Assumption 1 (so that $\mathbb{E}[\sum_\ell \|\mathbf{X} - g\| h_\ell^2] \lesssim \mathbb{E}[\sum_\ell h_\ell^2]$ and the cross-visit sums are controlled), and that $v \mapsto f_{\varepsilon|U,T,\mathbf{X},D \geq T}(v)$ is bounded with bounded derivative near 0. Then the population drift $R = M(\beta, m, g; \omega) + S_{\tau,\omega} \delta_\beta$ obeys*

$$|R| \leq C[(r_g + r_\omega)(r_m + r_\beta) + (r_m + r_\beta)^2],$$

i.e. the pure r_g , r_g^2 , r_ω , and $r_\omega r_g$ contributions are absent.

Proof. Split the integrand of (S2) by the two weight-side factors $\{\omega_0 \text{ or } \delta_\omega\} \times \{(\mathbf{X} - g_{\tau,\omega}) \text{ or } -\delta_g\}$, giving four pieces (A)–(D) with common scalar $\Psi = \sum_\ell \psi_\tau(\varepsilon_{0,\ell} - h_\ell)$. After the censoring identity and conditioning on $(U_\ell, T, \mathbf{X}, D \geq T)$, the bounded-derivative density gives the exact conditional-c.d.f. expansion

$$\mathbb{E}[\psi_\tau(\varepsilon_{0,\ell} - h_\ell) | U_\ell, T, \mathbf{X}, D \geq T] = -f_\ell(0) h_\ell + \rho_\ell, \quad |\rho_\ell| \leq C h_\ell^2. \quad (\text{S11})$$

(A) $\omega_0(\mathbf{X} - g_{\tau,\omega})$. Insert (S11): the δ_m -linear part is $D_m M[\delta_m] = 0$ (Lemma 2); the δ_β -linear part is $-S_{\tau,\omega} \delta_\beta$ (Corollary 1); the remainder is $O((r_m + r_\beta)^2)$. Thus (A) = $-S_{\tau,\omega} \delta_\beta + O((r_m + r_\beta)^2)$.

(B) $\omega_0(-\delta_g)$. At $h = 0$, $\mathbb{E}\{\psi_\tau(\varepsilon_{0,\ell}) | \cdot\} = 0$ (Lemma 1), so the pure- δ_g term vanishes ($D_g M[\delta_g] = 0$). The remaining piece uses (S11): (B) = $\mathbb{E}[\mathbf{1}\{T \leq D\} r_\omega \sum_\ell f_\ell(0) \delta_g h_\ell] + O(r_g(r_m + r_\beta)^2) = O(r_g(r_m + r_\beta))$ by Cauchy–Schwarz in $\nu_{\tau,\omega}$. **No r_g or r_g^2 term** (M is affine in g).

(C) $\delta_\omega(\mathbf{X} - g_{\tau,\omega})$. At $h = 0$ the conditional mean is 0, killing the pure- δ_ω term. With (S11), (C) = $-\mathbb{E}[\mathbf{1}\{T \leq D\} G_{C,0} \delta_\omega \sum_\ell f_\ell(0) \{\mathbf{X} - g_{\tau,\omega}\} h_\ell] + O(r_\omega(r_m + r_\beta)^2) = O(r_\omega(r_m + r_\beta))$ (this cross term need not vanish, as $g_{\tau,\omega}$ orthogonalizes against the r_ω -weighted measure, not the $G_{C,0} \delta_\omega$ -weighted one). **No pure r_ω term.**

(D) $\delta_\omega(-\delta_g)$. Both factors are perturbations; at $h = 0$ the conditional mean is 0, so (D) = $\mathbb{E}[\mathbf{1}\{T \leq D\} G_{C,0} \delta_\omega \sum_\ell f_\ell(0) \delta_g h_\ell] + \dots = O(r_\omega r_g(r_m + r_\beta))$, *third order*. **The pure $r_\omega r_g$ term is absent**; it appears only multiplied by h .

Summing (A)–(D) and subtracting $M(\beta_{\tau,0}, \eta_0) = 0$ (Proposition 1) gives $R = (B) + (C) + (D) + O((r_m + r_\beta)^2)$, hence the stated bound. For the CC member $\delta_\omega \equiv 0$ ($r_\omega = 0$); for the IPCW member $r_\omega \asymp \|\hat{G}_C - G_{C,0}\|_\infty / c_G^2$ under positivity. $\square$

Proposition 3 (Product form of the projected sieve bias). *Let $\mathcal{B}_J$ be the tensor-spline space, $g_{\tau,\omega,J} = \arg \min_{h \in \mathcal{B}_J^p} \mathbb{E}[\mathbf{1}\{T \leq D\} r_\omega \sum_\ell f_\ell(0) \|\mathbf{X} - h(U_\ell, T)\|^2]$ the finite-sieve $L_2(\nu_{\tau,\omega})$ projection of $g_{\tau,\omega}$, and $m_{\tau,J} = B^\top \gamma_{\tau,J}$ the spline component of the joint-sieve pseudo-truth θ_n° (Supplement II), with residual $r_{\tau,J} = m_{\tau,0} - m_{\tau,J}$. (The identity below holds for any (U, T) -measurable $r_{\tau,J}$; it is the pseudo-truth approximant that enters the actual bias.) Then*

$$\mathbb{E} \left[\mathbf{1}\{T \leq D\} r_\omega \sum_\ell f_\ell(0) \{\mathbf{X} - g_{\tau,\omega,J}\} r_{\tau,J}(U_\ell, T) \right] = \mathbb{E} \left[\mathbf{1}\{T \leq D\} r_\omega \sum_\ell f_\ell(0) \{g_{\tau,\omega} - g_{\tau,\omega,J}\} r_{\tau,J}(U_\ell, T) \right],$$

and consequently the projected approximation bias of the one-shot joint estimator obeys

$$|b_{K,\tau,\omega}| \leq \|S_{\tau,\omega,J}^{-1}\| \|g_{\tau,\omega} - g_{\tau,\omega,J}\|_{L_2(\nu_{\tau,\omega})} \|m_{\tau,0} - m_{\tau,J}\|_{L_2(\nu_{\tau,\omega})},$$

where $S_{\tau,\omega,J}$ is the finite-sieve information (its limit is $S_{\tau,\omega}$). The full pseudo-truth bias adds the square term of Remark 1.

Proof. Add and subtract $g_{\tau,\omega}$ inside the left-hand integrand: $\{\mathbf{X} - g_{\tau,\omega,J}\} = \{\mathbf{X} - g_{\tau,\omega}\} + \{g_{\tau,\omega} - g_{\tau,\omega,J}\}$. The term with $\{\mathbf{X} - g_{\tau,\omega}\}$ tested against the (U, T) -measurable direction $r_{\tau,J}$ equals $\int r_{\tau,J} (d\mu_{\tau,\omega} - g_{\tau,\omega} d\nu_{\tau,\omega}) = 0$ by the Radon–Nikodym characterization of $g_{\tau,\omega}$ (Lemma 2); note $r_{\tau,J}$ need *not* lie in $\mathcal{B}_J$, only be (U, T) -measurable. The stated identity follows. The bias bound is then Cauchy–Schwarz in $\nu_{\tau,\omega}$ on the right-hand inner product $\langle g_{\tau,\omega} - g_{\tau,\omega,J}, r_{\tau,J} \rangle_{\nu_{\tau,\omega}}$, times $\|S_{\tau,\omega}^{-1}\|$ from the β -block solve and uniform nonsingularity of $S_{\tau,\omega}$. $\square$

Remark 1 (Full pseudo-truth bias carries an additional square term). Proposition 3 bounds the *first-order* projected bias $b_{K,\tau,\omega}^{(1)}$. The full finite-sieve pseudo-truth offset $\beta_{\tau,J}^\circ - \beta_{\tau,0}$ also contains the second-order term from the conditional-c.d.f. nonlinearity (S11), $O(\|r_{\tau,J}\|^2)$. Hence the complete bound is

$$\|\beta_{\tau,J}^\circ - \beta_{\tau,0}\| \leq C(\|g_{\tau,\omega} - g_{\tau,\omega,J}\|_{L_2(\nu)} \|m_{\tau,0} - m_{\tau,J}\|_{L_2(\nu)} + \|m_{\tau,0} - m_{\tau,J}\|_{L_2(\nu)}^2),$$

so with $\|g - g_J\| = O(J^{-s_g})$, $\|m_0 - m_J\| = O(J^{-s_m})$ a centered root- n one-shot A-CV limit requires *both* $\sqrt{n} J^{-(s_g+s_m)} \rightarrow 0$ and $\sqrt{n} J^{-2s_m} \rightarrow 0$. The square term must *not* be dropped; it is what makes the one-shot A-CV requirement depend on s_m separately from the product $s_g + s_m$.

S1.0.4 High-level cross-fitted one-step theorem

We now state and prove asymptotic normality of (S7) under explicit nuisance-rate conditions, *without* reference to any concrete sieve rate. The first ingredient is an $L_2(P)$ score-reduction lemma that replaces a Donsker condition by cross-fitting and an L_1 -to- L_2 contraction special to the indicator score.

Lemma 7 (Cross-fitted L_2 score reduction). *Fix the fold- k nuisance estimates $\hat{\eta}^{(-k)}$, trained off fold k . Let $d_{i\ell} = \hat{m}^{(-k)}(U_{i\ell}, T_i) - m_0(U_{i\ell}, T_i) + \mathbf{X}_i^\top (\hat{\beta}_{\text{init}}^{(-k)} - \beta_{\tau,0})$. If $\|\hat{\omega}^{(-k)} - \omega_0\|_\infty + \|\hat{g}^{(-k)} - g_{\tau,\omega}\|_{L_2(\nu_{\tau,\omega})} = o_p(1)$ and the occupation-weighted mean of $|d_{i\ell}|$ is $o_p(1)$, then $\|\hat{\phi}^{(-k)} - \phi_{\tau,\omega}^0\|_{P,2} = o_p(1)$, and hence $\sqrt{n_k}(\mathbb{P}_{n,k} - P)\{\hat{\phi}^{(-k)} - \phi_{\tau,\omega}^0\} = o_p(1)$ conditionally on the training fold (no Donsker condition).*

Proof. Conditionally on the training fold, $\hat{\phi}^{(-k)}$ is a fixed function and fold- k subjects are i.i.d.; so by Chebyshev the conditional variance of $\sqrt{n_k}(\mathbb{P}_{n,k} - P)(\hat{\phi}^{(-k)} - \phi^0)$ is $\|\hat{\phi}^{(-k)} - \phi^0\|_{P,2}^2$, and it suffices to show the latter is $o_p(1)$. Decompose $\hat{\phi}^{(-k)} - \phi^0$ into (a) the weight/centering replacement $\{\hat{a}^{(-k)}(\mathbf{X} - \hat{g}^{(-k)}) - a_\omega(\mathbf{X} - g_{\tau,\omega})\}\psi_\tau(\varepsilon_0)$ and (b) the score-argument change $\hat{a}^{(-k)}(\mathbf{X} - \hat{g}^{(-k)})\{\psi_\tau(\varepsilon_0 - d) - \psi_\tau(\varepsilon_0)\}$. Term (a) is $O_p(r_\omega + r_g) = o_p(1)$ in $L_2(P)$ by boundedness of $\mathbf{X}, \omega, 1/G_C$ and $|\psi_\tau| \leq 1$. For term (b), the key contraction is that, conditionally on the design $Z = (U_\ell, T, \mathbf{X})$ and using $|\psi_\tau(\varepsilon_0 - d) - \psi_\tau(\varepsilon_0)| = |\mathbf{1}\{\varepsilon_0 \leq d\} - \mathbf{1}\{\varepsilon_0 \leq 0\}|$,

$$\mathbb{E}[\{\psi_\tau(\varepsilon_0 - d) - \psi_\tau(\varepsilon_0)\}^2 | Z] = \mathbb{P}\{0 < \varepsilon_0 \leq d | Z\} + \mathbb{P}\{d < \varepsilon_0 \leq 0 | Z\} \leq C|d|, \quad (\text{S12})$$

by the bounded conditional density. Term (b) is a *subject* sum $\sum_{\ell \leq M_i} q_{i\ell}$ with $q_{i\ell} = a_\omega(\mathbf{X} - \hat{g}^{(-k)})\{\psi_\tau(\varepsilon_0 - d) - \psi_\tau(\varepsilon_0)\}$, so its square has cross-visit terms; by Cauchy-Schwarz at the cluster level (Assumption 1) $(\sum_{\ell \leq M_i} q_{i\ell})^2 \leq M_i \sum_{\ell \leq M_i} q_{i\ell}^2$, whence, taking conditional expectation with (S12) and the boundedness of $\mathbf{X}, \hat{g}^{(-k)}, a_\omega$,

$$\|\text{term (b)}\|_{P,2}^2 \leq C' \mathbb{E} \left[M_i a_\omega^2 \sum_{\ell \leq M_i} \|\mathbf{X} - \hat{g}^{(-k)}\|^2 |d_{i\ell}| \right] = o_p(1),$$

since $\mathbb{E}[M_i^2] < \infty$, $\mathbf{X}, \hat{g}^{(-k)}$ are bounded, and the occupation-weighted mean of $|d_{i\ell}|$ is $o_p(1)$; the $L_2(\nu_{\tau,\omega})$ consistency of $\hat{g}^{(-k)}$ transfers to the score law $L_2(P)$ by the norm equivalence in Assumption 1. Summing (a) and (b) gives $\|\hat{\phi}^{(-k)} - \phi^0\|_{P,2} = o_p(1)$. $\square$

Theorem 2 (Cross-fitted one-step asymptotic normality, high-level). *Fix $\tau \in (0, 1)$ and a limiting weight ω_0 satisfying (S1). Suppose: (C1) the observed-data restriction (S9) holds (Lemma 1); (C2) S_{τ,ω_0} is nonsingular and, for some $\delta > 0$,*

$$\mathbb{E}\|\phi_{i,\tau,\omega_0}^0\|^{2+\delta} < \infty; \quad (\text{S13})$$

(C3) the conditional residual density is bounded away from 0 and ∞ and uniformly Lipschitz near 0; (C4) K fixed, $\min_k n_k/n \geq c_K > 0$; (C5) uniformly over folds, with $r_{d,k}^2 = \mathbb{E}[\mathbf{1}\{T \leq D\}r_{\omega_0} \sum_\ell f_\ell(0)d_{i\ell}^2]$, $r_{g,k} = \|\hat{g}^{(-k)} - g_{\tau,\omega_0}\|_{L_2(\nu)}$, and $r_{\omega,k} = \|\hat{\omega}^{(-k)} - \omega_0\|_\infty$,

$$r_{d,k} = o_p(1), \quad r_{g,k} + r_{\omega,k} = o_p(1), \quad (r_{g,k} + r_{\omega,k})r_{d,k} + r_{d,k}^2 = o_p(n^{-1/2});$$

(C6) $\|\hat{\phi}^{(-k)} - \phi^0\|_{P,2} = o_p(1)$ (Lemma 7); (C7) $\|\hat{S}_k - S_{\tau,\omega_0}\|_{\text{op}} = o_p(1)$ and $\|\hat{S}_k - S_{\tau,\omega_0}\|_{\text{op}} \sqrt{n} \|\hat{\beta}_{\text{init}}^{(-k)} - \beta_{\tau,0}\| = o_p(1)$. Then

$$\sqrt{n}(\tilde{\beta}_{\tau,\omega_0} - \beta_{\tau,0}) = S_{\tau,\omega_0}^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \phi_{i,\tau,\omega_0}^0 + o_p(1) \rightsquigarrow \mathcal{N}(0, \Sigma_{\tau,\omega_0}),$$

$\Sigma_{\tau,\omega_0} = S_{\tau,\omega_0}^{-1} \mathbb{E}(\phi_{i,\tau,\omega_0}^0 \phi_{i,\tau,\omega_0}^{0\top}) S_{\tau,\omega_0}^{-\top}$. For CC, $r_{\omega,k} = 0$; for IPCW, $r_{\omega,k} = O_p(\|\hat{G}_C - G_{C,0}\|_\infty)$.

Proof. Fix fold k and suppress k . Decompose $\mathbb{P}_{n,k} \hat{\phi}(\hat{\beta}_{\text{init}}) = (\mathbb{P}_{n,k} - P)\phi^0 + (\mathbb{P}_{n,k} - P)(\hat{\phi} - \phi^0) + P\hat{\phi}(\hat{\beta}_{\text{init}})$. The middle term is $o_p(n^{-1/2})$ conditionally on the training fold by (C6) and Lemma 7. The last term is the population drift $P\hat{\phi}(\hat{\beta}_{\text{init}}) = M(\hat{\beta}_{\text{init}}, \hat{m}, \hat{g}, \hat{\omega})$; by Lemma 6,

$$P\hat{\phi}(\hat{\beta}_{\text{init}}) = -S_{\tau,\omega_0}(\hat{\beta}_{\text{init}} - \beta_{\tau,0}) + R_k, \quad |R_k| \leq C[(r_{g,k} + r_{\omega,k})r_{d,k} + r_{d,k}^2] = o_p(n^{-1/2})$$

by (C5). Therefore $\sqrt{n_k} \mathbb{P}_{n,k} \hat{\phi}(\hat{\beta}_{\text{init}}) = \sqrt{n_k} (\mathbb{P}_{n,k} - P) \phi^0 - S_{\tau, \omega_0} \sqrt{n_k} (\hat{\beta}_{\text{init}} - \beta_{\tau, 0}) + o_p(1)$. Substituting into the fold bracket of (S7) and replacing $\hat{S}_k^{-1}$ by S_{τ, ω_0}^{-1} ,

$$\sqrt{n_k} [\hat{\beta}_{\text{init}} - \beta_{\tau, 0} + \hat{S}_k^{-1} \mathbb{P}_{n,k} \hat{\phi}(\hat{\beta}_{\text{init}})] = S_{\tau, \omega_0}^{-1} \sqrt{n_k} (\mathbb{P}_{n,k} - P) \phi^0 + e_k + o_p(1),$$

where the initial-estimator first-order term cancels exactly and the substitution error is $e_k = (\hat{S}_k^{-1} - S_{\tau, \omega_0}^{-1}) S_{\tau, \omega_0} \sqrt{n_k} (\hat{\beta}_{\text{init}} - \beta_{\tau, 0})$. By (C7) and the identity $\hat{S}_k^{-1} - S^{-1} = -\hat{S}_k^{-1} (\hat{S}_k - S) S^{-1}$, $\|e_k\| \leq \|\hat{S}_k^{-1}\| \|\hat{S}_k - S_{\tau, \omega_0}\|_{\text{op}} \sqrt{n_k} \|\hat{\beta}_{\text{init}} - \beta_{\tau, 0}\| = o_p(1)$. Aggregating with the size weights and using $\sum_k (n_k/n) \mathbb{P}_{n,k} = \mathbb{P}_n$,

$$\sqrt{n} (\tilde{\beta}_{\tau, \omega_0} - \beta_{\tau, 0}) = S_{\tau, \omega_0}^{-1} \sqrt{n} (\mathbb{P}_n - P) \phi^0 + o_p(1).$$

The subjects $\phi_{i, \tau, \omega_0}^0$ are i.i.d. under P_S with mean 0 (Proposition 1) and finite $2 + \delta$ moment (C2); the Lindeberg CLT and Slutsky give the stated limit. Note the quadratic remainder $\|\hat{\beta}_{\text{init}} - \beta_{\tau, 0}\|^2$ does not cancel, so $\hat{\beta}_{\text{init}}$ must satisfy $\sqrt{n} r_d^2 = o_p(1)$, which (C5) supplies; mere $o_p(1)$ consistency of $\hat{\beta}_{\text{init}}$ is insufficient. $\square$

Corollary 2 (Cross-fitted sandwich consistency). *Let $\hat{\zeta}_i = \hat{S}_{k(i)}^{-1} \hat{\phi}_i^{(-k(i))} (\hat{\beta}_{\text{init}}^{(-k(i))})$, $\bar{\zeta} = n^{-1} \sum_i \hat{\zeta}_i$, and $\hat{\Sigma} = n^{-1} \sum_i (\hat{\zeta}_i - \bar{\zeta})(\hat{\zeta}_i - \bar{\zeta})^\top$. Under (C2), (C6), (C7), $\hat{\Sigma} \rightarrow_p \Sigma_{\tau, \omega_0}$.*

Proof. Across folds the $\hat{\zeta}_i$ are *not* globally i.i.d. (fold- k nuisances depend on the other folds' data), so we decompose by fold: $\hat{\Sigma} + \bar{\zeta}^{\otimes 2} = \sum_{k=1}^K \frac{n_k}{n} \mathbb{P}_{n,k} [\hat{\zeta}_k^{\otimes 2}]$, with $\hat{\zeta}_k = \hat{S}_k^{-1} \hat{\phi}^{(-k)}$. Fix k and condition on the training data (the complement of fold k): then $\hat{S}_k^{-1}$ and the function $\hat{\phi}^{(-k)}$ are *fixed*, and fold- k subjects are i.i.d., so by the conditional LLN $\mathbb{P}_{n,k} [\hat{\zeta}_k^{\otimes 2}] - P[\hat{\zeta}_k^{\otimes 2}] = o_p(1)$ (finite second moment from C2 and boundedness). Next $P[\hat{\zeta}_k^{\otimes 2}] \rightarrow_p S_{\tau, \omega_0}^{-1} \mathbb{E}(\phi^0 \phi^{0\top}) S_{\tau, \omega_0}^{-\top}$ because $\hat{S}_k^{-1} \rightarrow_p S_{\tau, \omega_0}^{-1}$ (C7) and $P\|\hat{\zeta}_k - S_{\tau, \omega_0}^{-1} \phi^0\|^2 \leq \|\hat{S}_k^{-1}\|^2 \|\hat{\phi}^{(-k)} - \phi^0\|_{P, 2}^2 + \|\hat{S}_k^{-1} - S_{\tau, \omega_0}^{-1}\|_{\text{op}}^2 \mathbb{E}\|\phi^0\|^2 = o_p(1)$ (C6, C7, C2). Aggregating over the fixed K folds with $\min_k n_k/n \geq c_K$, and $\bar{\zeta} \rightarrow_p 0$, gives $\hat{\Sigma} \rightarrow_p \Sigma_{\tau, \omega_0}$. $\square$

S1.0.5 What is deferred, and the two routes

Theorem 2 assumes (C5)–(C7) as *high-level* rates. That a concrete tensor-B-spline / quantile-spacing estimator attains them—in particular the parametric block rate $r_d = o_p(n^{-1/4})$ and the bread consistency at the required order—is proved in supplement II via a clustered weighted growing-sieve Bahadur representation, whose admissible window is pinned by a density-weighted Gram concentration that *exploits the local support* of the tensor splines. We do not claim those empirical-process inputs here.

Remark 2 (Two routes). **Route A (default point estimator + inference): the A-CV joint estimator** optimizes the check loss directly, forms neither $\hat{f}$ nor an explicit $\hat{g}$, and is reported with the subject bootstrap; its *one-shot* analytic root- n theory needs $\sqrt{n} b_{K, \tau, \omega} \rightarrow 0$, controlled in product form by Proposition 3. **Route B (explicit orthogonal estimator): the cross-fitted one-step B-OS-CF** of (S7) has the clean orthogonal score and, under (C5)–(C7), removes the first-order projected bias $b_{K, \tau, \omega}$ to deliver the limit of Theorem 2. The bootstrap propagates sampling and tuning variability but does *not* remove a first-order sieve bias when $\sqrt{n} b_{K, \tau, \omega} \not\rightarrow 0$; that is the role of the one-step correction.

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S1.B Censoring orthogonality of the IPW quantile score

Summary / correction. An earlier note claimed that plugging in an estimated censoring distribution $\hat{G}_C$ adds a nonzero first-order term χ^C to the influence function of $\hat{\beta}_\tau$. **That claim was wrong.** The IPW estimating function built from a *conditional*-quantile restriction is Neyman-orthogonal to the censoring nuisance, so $\chi^C = 0$ at first order; $\hat{G}_C$ enters only through a second-order nuisance-product remainder. This note proves the orthogonality and explains exactly where the earlier derivation stopped one step too early. The simulation corroborates it: the frozen- G_C and full-refit bootstraps coincide.

S1.0.6 Setup

With $H = (A, \mathbf{X})$, $W = T - A$, $G_{C,0}(w|H) = \mathbb{P}(C \geq w|H)$, $\Delta^T = \mathbf{1}\{T \leq D, C \geq W\}$, $\psi_\tau(r) = \tau - \mathbf{1}\{r \leq 0\}$, $\varepsilon_{i\ell,\tau} = L_{i\ell} - m_{\tau,0}(U_{i\ell}, T_i) - \beta_{\tau,0}^\top \mathbf{X}_i$, and the centered, summed score

$$\xi_{i,\tau} = \sum_{\ell} \{\mathbf{X}_i - g_{\tau,0}(U_{i\ell}, T_i)\} \psi_\tau(\varepsilon_{i\ell,\tau}),$$

the IPW population moment is $\Psi(G_C) = \mathbb{E}[(\Delta^T/G_C(W|H))\xi_\tau]$. We maintain (i) the conditional-quantile model $\mathbb{E}[\psi_\tau(\varepsilon_{i\ell,\tau}) | U_{i\ell}, T_i, \mathbf{X}_i, D_i \geq T_i] = 0$, (ii) independent censoring $C \perp \{Y(\cdot), T, D\} | H$, and (iii) noninformative visits.

S1.0.7 The vanishing of the path derivative

Perturb $G_{C,\epsilon} = G_{C,0} + \epsilon h$. Differentiating the weight $1/G_C$,

$$\dot{\Psi}_{G_C}[h] = -\mathbb{E}\left[\Delta^T \xi_\tau \frac{h(W|H)}{G_{C,0}(W|H)^2}\right]. \quad (\text{S14})$$

The earlier note stopped here and declared (S14) nonzero. Continue. Condition on the full data $(T, D, A, \mathbf{X}, Y, \{U_\ell\})$, under which only C is random; by independent censoring $\mathbb{E}[\Delta^T | \text{full data}] = \mathbf{1}\{T \leq D\} G_{C,0}(W|H)$, so

$$\dot{\Psi}_{G_C}[h] = -\mathbb{E}\left[\mathbf{1}\{T \leq D\} \xi_\tau \frac{h(W|H)}{G_{C,0}(W|H)}\right]. \quad (\text{S15})$$

Now use the conditional-quantile restriction. Since $\{\mathbf{X}_i - g_{\tau,0}(U_{i\ell}, T_i)\}$ is a function of $(\mathbf{X}_i, U_{i\ell}, T_i)$,

$$\mathbb{E}[\{\mathbf{X}_i - g_{\tau,0}(U_{i\ell}, T_i)\}\psi_\tau(\varepsilon_{i\ell,\tau}) \mid U_{i\ell}, T_i, \mathbf{X}_i, D_i \geq T_i] = 0,$$

and under noninformative visits, summing over ℓ ,

$$\mathbb{E}[\xi_{i,\tau} \mid T_i, A_i, \mathbf{X}_i, D_i \geq T_i, \{U_{i\ell}\}] = 0. \quad (\text{S16})$$

The factor $h(W \mid H)/G_{C,0}(W \mid H)$ and $\mathbf{1}\{T \leq D\}$ are measurable with respect to the conditioning σ -field in (S16). By the tower property, (S15) becomes

$$\dot{\Psi}_{G_C}[h] = -\mathbb{E}\left[\mathbf{1}\{T \leq D\} \frac{h(W \mid H)}{G_{C,0}(W \mid H)} \mathbb{E}\{\xi_\tau \mid T, A, \mathbf{X}, D \geq T, \{U_\ell\}\}\right] = 0.$$

Lemma 8. *Under the conditional-quantile model, independent censoring, and noninformative visits, $\dot{\Psi}_{G_C}[h] = 0$ for every admissible direction h . Hence the plug-in $\hat{G}_C$ contributes no first-order term: $\chi_{i,\tau}^C = \dot{\Psi}_{G_C}[\text{IF}_{G_C,i}] = 0$.*

S1.0.8 Why ordinary IPW *does* have a censoring term — and we do not

For a *marginal* target (e.g. $\mathbb{E}Y$ via $(\Delta/G_C)(Y - \mu)$) one has $\mathbb{E}(Y - \mu \mid H) \neq 0$, so (S15) does not vanish and the propensity/censoring model enters the first-order influence function. Our score rests on the stronger *conditional* restriction $\mathbb{E}[\psi_\tau(\varepsilon) \mid U, T, \mathbf{X}] = 0$; multiplying a conditionally mean-zero quantity by any weight depending only on the conditioning variables keeps it mean zero. IPCW here only reweights which onset subjects are seen and how much they count—it shifts the design measure (hence efficiency and the finite-sieve pseudo-true value) but not the first-order limit.

S1.0.9 Consequences

- The feasible and oracle-weight expansions share the leading influence function $S_\tau^{-1} n^{-1/2} \sum_i \phi_{i,\tau}^0$, with no χ^C .
- $\hat{G}_C$ enters only at second order, through a nuisance-product remainder of order $\|\hat{G}_C - G_{C,0}\| \cdot \|\hat{m}_\tau - m_{\tau,0}\|$ (and likewise with $\hat{g}_\tau$), controlled by cross-fitting and the product-rate condition.
- Consequently IPCW is *not* required for identification: the unweighted complete-event score $\mathbb{E}[\Delta^T \xi_\tau] = \mathbb{E}[\mathbf{1}\{T \leq D\} G_{C,0} \mathbb{E}(\xi_\tau \mid \cdot)] = 0$ also identifies $(m_{\tau,0}, \beta_{\tau,0})$; it is a design/efficiency choice.
- The orthogonality can fail (so $\chi^C \neq 0$) if: censoring depends on the unobserved marker but G_C conditions only on $(A, \mathbf{X})$; the conditional quantile model is misspecified; visits are informative; the target is a marginal quantile; or one targets a fixed misspecified finite-sieve pseudo-parameter.

S1.C Clustered weighted growing-sieve Bahadur representation

Purpose. Supplement I’s high-level one-step theorem assumes nuisance-rate conditions (C5–C7 there). This note *proves* that the first-step joint-sieve quantile-regression estimator attains them, supplying the parametric Bahadur representation $\hat{\beta}_{\text{init}} - \beta_{\tau,0} = S^{-1} \mathbb{P}_n \phi^0 + b_K + o_p(n^{-1/2})$. The two inputs flagged “open” in the earlier draft—a clustered Knight maximal inequality and a density-weighted empirical-Gram concentration—are established here (Lemmas 9 and 11). The single technical point that makes the admissible sieve window non-vacuous for smoothness $s > 1$ is that the tensor B-spline Gram is *locally supported*; ignoring this forces $J_n = o(n^{1/6})$ and $s > 3/2$. We prove the oracle-weight result and then add the estimated weight as a perturbation (§S1.0.14).

S1.0.10 Setup and regularity

Let $O_1, \dots, O_n$ be i.i.d. subject-level observations under the retained-sample law P_S ; within-subject measurements may be arbitrarily dependent. Stack $\boldsymbol{\theta} = (\boldsymbol{\gamma}^\top, \boldsymbol{\beta}^\top)^\top \in \mathbb{R}^{d_n}$, $d_n = K_n + p$, $K_n = J_n^2$, and the design vector $\mathbf{Z}_{i\ell} = (B_n(U_{i\ell}, T_i)^\top, \mathbf{X}_i^\top)^\top$ with B_n the normalized tensor cubic B-spline basis. For a fixed bounded positive observed-event weight ω_0 write $a_i = \Delta_i^T \omega_0(T_i, A_i, \mathbf{X}_i) \in [0, C_\omega]$. The estimator solves the convex program

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \mathbb{P}_n q_{i,n}(\boldsymbol{\theta}) + \frac{1}{2} \boldsymbol{\theta}^\top \Lambda_n \boldsymbol{\theta}, \quad q_{i,n}(\boldsymbol{\theta}) = a_i \sum_{\ell=1}^{M_i} \rho_\tau(L_{i\ell} - \mathbf{Z}_{i\ell}^\top \boldsymbol{\theta}),$$

$\Lambda_n = \text{diag}(\lambda_n P_n, 0)$ the (optional) roughness penalty on the spline block. Let $\boldsymbol{\theta}_n^\circ = \arg \min_{\boldsymbol{\theta}} P q_{i,n}(\boldsymbol{\theta})$ be the *population sieve pseudo-truth* (so the population score vanishes there) and $\varepsilon_{i\ell} = L_{i\ell} - \mathbf{Z}_{i\ell}^\top \boldsymbol{\theta}_n^\circ$.

Condition 1 (Regularity). (A1) O_i i.i.d. under P_S . (A2) $0 < c_\omega \leq \omega_0 \leq C_\omega < \infty$. (A3) the conditional density $f_{i\ell,n}$ of $\varepsilon_{i\ell,n}$ given the design satisfies $0 < c_f \leq f_{i\ell,n}(0) \leq C_f$ and $|f_{i\ell,n}(v) - f_{i\ell,n}(0)| \leq L|v|$ near 0. (A4) (local support). The normalized tensor B-splines have local support: at each (u, t) at most q_0 basis functions are nonzero, the overlap graph has uniformly bounded degree $\bar{D}$, and $\zeta_n := \sup_{u,t} \|B_n(u, t)\|_2 \asymp \sqrt{K_n}$. (A5) the population density-weighted Gram $H_n = \mathbb{E}[a_i \sum_{\ell} f_{i\ell,n}(0) \mathbf{Z}_{i\ell} \mathbf{Z}_{i\ell}^\top]$ has eigenvalues in $[\underline{c}, \bar{c}]$, $0 < \underline{c} \leq \bar{c} < \infty$, uniformly in n . (A6) (cluster envelope). $M_i(1 + \|\mathbf{X}_i\|)^2$ is sub-exponential: $\|M_i(1 + \|\mathbf{X}_i\|)^2\|_{\psi_1} \leq C_M < \infty$. (A7) $\|\Lambda_n\|_{\text{op}} = o(1)$.

Assumption (A6) is stronger than the $2+\delta$ moment that suffices for the fixed-dimensional CLT of supplement I; it is needed for *growing-dimensional* operator-norm concentration. A finite-moment variant (q th moment, $q > 4$, with envelope truncation at a slowly growing level b_n whose discarded tail is controlled by Markov) yields the same conclusions at the cost of an extra b_n factor in the sieve-growth condition; we state the clean sub-exponential version and note the truncation route in Remark 4.

S1.0.11 Density-weighted Gram concentration (exploiting local support)

Define $\hat{H}_n = \mathbb{P}_n[a_i \sum_{\ell} f_{i\ell,n}(0) \mathbf{Z}_{i\ell} \mathbf{Z}_{i\ell}^\top]$.

Lemma 9 (Clustered Gram concentration). *Under Condition 1,*

$$\|\hat{H}_n - H_n\|_{\text{op}} = O_p(\delta_{H,n}), \quad \delta_{H,n} = \sqrt{\frac{K_n \log K_n}{n}} + \frac{K_n \log K_n}{n}.$$

Hence if $K_n \log K_n / n \rightarrow 0$ then $\|\hat{H}_n - H_n\|_{\text{op}} = o_p(1)$ and $\lambda_{\min}(\hat{H}_n) \geq \underline{c}/2$ with probability $\rightarrow 1$.

Proof. Partition $\hat{H}_n - H_n$ into the spline–spline, spline–covariate, and covariate–covariate blocks; bound the operator norm by the sum of the three block operator norms.

Spline–spline block. Its (j, k) entry is the centered subject average of $Y_{i,jk} = a_i \sum_{\ell} f_{i\ell,n}(0) B_{j,n}(U_{i\ell}, T_i) B_{k,n}(U_{i\ell}, T_i)$. By local support (A4), $B_{j,n} B_{k,n} \equiv 0$ unless supports j, k overlap; the number of overlapping ordered pairs is $O(K_n \bar{D}) = O(K_n)$, and each row has at most $q_0 \bar{D} = O(1)$ nonzero entries. For an overlapping pair, the normalized splines satisfy $|B_{j,n}| \lesssim \zeta_n \asymp \sqrt{K_n}$ on their (shared) support, whose probability under the design is $O(K_n^{-1})$ (since the occupation density is bounded, A5, and there are K_n cells); together with boundedness of $a_i f$, the per-pair conditional second moment satisfies $\mathbb{E}[Y_{i,jk}^2] \lesssim (\zeta_n^2)^2 \cdot K_n^{-1} \cdot \mathbb{E}[a_i^2 M_i^2] = O(K_n) \cdot \mathbb{E}[a_i^2 M_i^2]$, finite by (A6), and the envelope is

$|Y_{i,jk}| \lesssim a_i M_i \zeta_n^2 = O(M_i K_n)$. Bernstein's inequality for the i.i.d. subject sum gives, for each fixed pair,

$$\mathbb{P}(|(\mathbb{P}_n - P)Y_{i,jk}| > t) \leq 2 \exp\left(-\frac{cnt^2}{K_n + M_* K_n t}\right), \quad M_* = \|M_i\|_{\psi_1},$$

using $\text{Var}(Y_{i,jk}) = O(K_n)$ and the sub-exponential envelope (A6). A union bound over the $O(K_n)$ nonzero pairs, with $t \asymp \sqrt{K_n \log K_n/n} + K_n \log K_n/n$, yields $\max_{(j,k): B_j B_k \neq 0} |(\mathbb{P}_n - P)Y_{i,jk}| = O_p(\delta_{H,n})$. Because the overlap graph has bounded degree (A4), the spline-spline block is banded with $O(1)$ nonzeros per row, so its operator norm is at most the maximum absolute row sum $\leq q_0 \bar{D} \cdot \max_{(j,k)} |(\mathbb{P}_n - P)Y_{i,jk}| = O_p(\delta_{H,n})$.

Spline-covariate block. It has $K_n p$ entries (p fixed). The same per-entry Bernstein bound applies; each spline column overlaps $O(1)$ rows, so the block operator norm (bounded by $\sqrt{(\#\text{nonzero rows}) \times \text{max entry}}$, again $O(1)$ nonzeros per spline row) is $O_p(\delta_{H,n})$, using the finite moments of $\mathbf{X}_i$ from (A6).

Covariate-covariate block. Fixed $p \times p$; a standard $O_p(n^{-1/2})$, dominated by $\delta_{H,n}$.

Summing the three blocks gives $\|\hat{H}_n - H_n\|_{\text{op}} = O_p(\delta_{H,n})$. The eigenvalue statement follows from Weyl's inequality and (A5). Within-subject dependence is irrelevant: $Y_{i,jk}$ is one cluster-level summand and independence is used only across i . $\square$

S1.0.12 Local Knight expansion and clustered maximal inequality

For $h \in \mathbb{R}^{d_n}$ define the per-subject Knight remainder

$$\mathcal{R}_{i,n}(h) = a_i \sum_{\ell} \left[\rho_{\tau}(\varepsilon_{i\ell,n} - \mathbf{Z}_{i\ell}^{\top} h) - \rho_{\tau}(\varepsilon_{i\ell,n}) + \psi_{\tau}(\varepsilon_{i\ell,n}) \mathbf{Z}_{i\ell}^{\top} h \right] = a_i \sum_{\ell} \int_0^{\mathbf{Z}_{i\ell}^{\top} h} \{ \mathbf{1}\{\varepsilon_{i\ell,n} \leq s\} - \mathbf{1}\{\varepsilon_{i\ell,n} \leq 0\} \} ds,$$

the last equality being Knight's identity (no differentiability used).

Lemma 10 (Population quadratic expansion). *Under Condition 1, uniformly for $\|h\| \leq r$ with $\zeta_n r = o(1)$,*

$$P \mathcal{R}_{i,n}(h) = \frac{1}{2} h^{\top} H_n h + O(\zeta_n \|h\|_{H_n}^2 \cdot \|h\|).$$

Proof. Condition on the full design $\mathcal{Z}_i$. For scalar $v = \mathbf{Z}_{i\ell}^{\top} h$, $\mathbb{E}\{\int_0^v (\mathbf{1}\{\varepsilon \leq s\} - \mathbf{1}\{\varepsilon \leq 0\}) ds \mid \mathcal{Z}_i\} = \int_0^v \{F_{i\ell,n}(s) - F_{i\ell,n}(0)\} ds$. By the local Lipschitz density (A3), $F_{i\ell,n}(s) - F_{i\ell,n}(0) = f_{i\ell,n}(0)s + O(s^2)$, so the integral is $\frac{1}{2} f_{i\ell,n}(0)v^2 + O(|v|^3)$. Summing over visits and weighting by a_i , the leading term is $\frac{1}{2} h^{\top} H_n h$. The cubic remainder obeys $\sum_{\ell} |\mathbf{Z}_{i\ell}^{\top} h|^3 \leq (\sup_{\ell} |\mathbf{Z}_{i\ell}^{\top} h|) \sum_{\ell} (\mathbf{Z}_{i\ell}^{\top} h)^2 \leq \zeta_n \|h\| \cdot \sum_{\ell} (\mathbf{Z}_{i\ell}^{\top} h)^2$, whose a -weighted expectation is $O(\zeta_n \|h\| \cdot \|h\|_{H_n}^2)$ by (A5). No within-subject independence is used (expectation is taken of the cluster sum). $\square$

Lemma 11 (Clustered Knight maximal inequality). *Under Condition 1, for a deterministic $r_n \downarrow 0$ with $\zeta_n r_n = o(1)$,*

$$\sup_{\|h\| \leq r_n} |(\mathbb{P}_n - P) \mathcal{R}_{i,n}(h)| = O_p\left(\sqrt{\frac{d_n \zeta_n r_n^3 \log n}{n}} + \frac{d_n \zeta_n r_n \log n}{n}\right).$$

In particular, if $r_n \asymp \sqrt{d_n/n}$ and $d_n^2(\log n)^c/n \rightarrow 0$ for a fixed c , the supremum is $o_p(r_n^2)$.

Proof. The class $\mathcal{F}_{n,r} = \{\mathcal{R}_{i,n}(h) : \|h\| \leq r_n\}$ is generated by the linear indices $\mathbf{Z}_{i\ell}^{\top} h$, threshold indicators, integration, and a finite per-subject sum; it is a VC-type class with envelope

$F_i \lesssim a_i M_i \zeta_n r_n$ (since $|\mathcal{R}_{i,n}(h)| \leq a_i \sum_{\ell} |\mathbf{Z}_{i\ell}^\top h| \mathbf{1}\{|\varepsilon_{i\ell,n}| \leq |\mathbf{Z}_{i\ell}^\top h|\} \leq a_i M_i \zeta_n r_n$) and uniform entropy $\log N(\epsilon \|F\|_{P,2}, \mathcal{F}_{n,r}, L_2(P)) \lesssim d_n \log(C/\epsilon)$, because the h -dependence enters through a d_n -dimensional linear index and the maps $v \mapsto \int_0^v (\mathbf{1}\{\varepsilon \leq s\} - \mathbf{1}\{\varepsilon \leq 0\}) ds$ are monotone Lipschitz of the index (a bounded-variation composition preserves the VC-subgraph property up to a constant). The *local variance* is sharper than the squared envelope: since the integrand is nonzero only on $\{|\varepsilon_{i\ell,n}| \leq |\mathbf{Z}_{i\ell}^\top h|\}$, whose conditional probability is $O(|\mathbf{Z}_{i\ell}^\top h|)$ by (A3),

$$\sup_{\|h\| \leq r_n} P\{\mathcal{R}_{i,n}(h)^2\} \lesssim \mathbb{E} \left[a_i^2 \left(\sum_{\ell} |\mathbf{Z}_{i\ell}^\top h| \mathbf{1}\{|\varepsilon_{i\ell,n}| \leq |\mathbf{Z}_{i\ell}^\top h|\} \right)^2 \right] \lesssim \zeta_n r_n^3,$$

using $|\mathbf{Z}^\top h| \leq \zeta_n r_n$ and the a -weighted second moment of the visit sum (A6). Apply a Bernstein-type maximal inequality for i.i.d. subject-level summands over a VC-type class (e.g. the bounded-variance empirical-process bound of van der Vaart–Wellner, Thm. 2.14.1, with the local variance $\sigma^2 \asymp \zeta_n r_n^3$ and the sub-exponential envelope (A6) controlling the higher-order term): the bound is $\lesssim \sqrt{d_n \sigma^2 \log n/n} + d_n \|F\|_{\psi_1} \log n/n$, which is the stated rate. The two prerequisites of that inequality—finite uniform entropy integral and a finite (here sub-exponential) envelope—are verified above; the sub-exponential envelope is exactly where (A6) is used and where a mere $2+\delta$ moment would not suffice. Plugging $r_n \asymp \sqrt{d_n/n}$ gives $\sqrt{d_n^2 \zeta_n (d_n/n)^{3/2} \log n/n} \asymp r_n^2 \cdot \sqrt{d_n^2 (\log n)/n \cdot \zeta_n / \sqrt{d_n}}$; since $\zeta_n \asymp \sqrt{K_n} \lesssim \sqrt{d_n}$, the bracket is $\sqrt{d_n^2 \log n/n} = o(1)$ under $d_n^2 (\log n)^c/n \rightarrow 0$, so the supremum is $o_p(r_n^2)$. $\square$

S1.0.13 Localization and profiled Bahadur representation

Let $G_n = \mathbb{P}_n[a_i \sum_{\ell} \mathbf{Z}_{i\ell} \psi_{\tau}(\varepsilon_{i\ell,n})]$; since $\boldsymbol{\theta}_n^\circ$ is the population sieve minimizer, $PG_n = 0$ and (bounded weights, A6) $\|G_n\| = O_p(\sqrt{d_n/n})$.

Lemma 12 (Localization). *Set $r_n = C\sqrt{d_n/n}$. If $\delta_{H,n} = o(1)$, $\zeta_n r_n = o(1)$, and Lemma 11 gives $o_p(r_n^2)$, then $\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_n^\circ\| = O_p(r_n + \|\Lambda_n \boldsymbol{\theta}_n^\circ\|)$.*

Proof. For $\|h\| = Mr_n$, Knight’s identity and Lemma 10 give $Q_n(\boldsymbol{\theta}_n^\circ + h) - Q_n(\boldsymbol{\theta}_n^\circ) = -h^\top G_n + \frac{1}{2} h^\top H_n h + (\mathbb{P}_n - P)\mathcal{R}_{i,n}(h) + \text{penalty}$. The linear term is $O_p(r_n \|h\|)$; by (A5) the quadratic dominates from below, $\frac{1}{2} h^\top H_n h \geq \frac{\epsilon}{2} \|h\|^2$; the empirical remainder is $o_p(\|h\|^2)$ (Lemma 11); the penalty contributes $O(\|\Lambda_n\| \|h\| (\|\boldsymbol{\theta}_n^\circ\| + \|h\|))$, lower order by (A7). For M large the objective is positive on $\{\|h\| = Mr_n\}$ w.p. $\rightarrow 1$; by convexity of $\boldsymbol{\theta} \mapsto Q_n(\boldsymbol{\theta})$ the minimizer lies inside the ball. The penalty bias adds $\|\Lambda_n \boldsymbol{\theta}_n^\circ\|$. $\square$

Lemma 13 (Profiled Bahadur representation). *Partition $H_n = \begin{pmatrix} A_n & C_n \\ C_n^\top & D_n \end{pmatrix}$, $S_n = D_n - C_n^\top A_n^{-1} C_n \rightarrow S_{\tau, \omega_0}$, and define the finite-sieve projection $g_{\tau, \omega_0, n}(u, t) = C_n^\top A_n^{-1} B_n(u, t)$ (so $g_{\tau, \omega_0, n} \rightarrow g_{\tau, \omega_0}$, *supplement I*). If, in addition to Lemmas 9–12, $\delta_{H,n} r_n = o(n^{-1/2})$, then*

$$\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}_n^\circ = S_n^{-1} \mathbb{P}_n \left[a_i \sum_{\ell} \{\mathbf{X}_i - g_{\tau, \omega_0, n}(U_{i\ell}, T_i)\} \psi_{\tau}(\varepsilon_{i\ell,n}) \right] + b_{\lambda,n} + o_p(n^{-1/2}),$$

$b_{\lambda,n}$ the profiled penalty bias.

Proof. The local subgradient (normal) equation reads $0 \in -G_n + \hat{H}_n(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_n^\circ) + \Lambda_n \hat{\boldsymbol{\theta}} + R_{n,sc}$, where the nonsmooth score-increment remainder $R_{n,sc}$ is the gradient analogue of $\mathcal{R}_{i,n}$, namely the centered process $\mathbf{Z}_{i\ell} \{\mathbf{1}\{\varepsilon_{i\ell,n} \leq \mathbf{Z}_{i\ell}^\top h\} - \mathbf{1}\{\varepsilon_{i\ell,n} \leq 0\}\}$, controlled to $o_p(n^{-1/2})$ on the localization ball $\|h\| \leq r_n$ by the same VC-type maximal inequality as Lemma 11 (its gradient version, identical entropy and envelope, local variance now $O(\zeta_n r_n)$). Replacing the empirical curvature $\hat{H}_n$ by H_n costs

$\|\widehat{H}_n - H_n\|_{\text{op}} \|\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_n^\circ\| = O_p(\delta_{H,n} r_n) = o_p(n^{-1/2})$ by hypothesis. Solving the two block equations, the $\boldsymbol{\beta}$ -block is the Schur complement $\widehat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}_n^\circ = S_n^{-1} \{G_{\beta,n} - C_n^\top A_n^{-1} G_{\gamma,n}\} + b_{\lambda,n} + o_p(n^{-1/2})$, and the combination $G_{\beta,n} - C_n^\top A_n^{-1} G_{\gamma,n}$ equals $\mathbb{P}_n[a_i \sum_\ell \{\mathbf{X}_i - g_{\tau,\omega_0,n}\} \psi_\tau(\varepsilon_{i\ell,n})]$ by the definition of $g_{\tau,\omega_0,n}$. Nonsingularity of A_n, S_n holds w.p. $\rightarrow 1$ by Lemma 9 and (A5). $\square$

S1.0.14 From the sieve pseudo-truth to the target; estimated weights

Adding the projected approximation bias of supplement I, Proposition 2 (product form), gives

$$\widehat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}_{\tau,0} = S_{\tau,\omega_0}^{-1} \mathbb{P}_n \phi_{\tau,\omega_0}^0 + b_{K,\tau,\omega_0} + b_{\lambda,n} + o_p(n^{-1/2}), \quad |b_{K,\tau,\omega_0}| \leq C \|g - g_n\|_{L_2(\nu)} \|m_0 - m_n\|_{L_2(\nu)} = O(J_n^{-(s_g + s_m)}),$$

the last equality being the *product* form of supplement I, Proposition 2. This is exactly the parametric representation feeding supplement I's C5: even the crude single-factor bound $|b_K| = O(J_n^{-s_m})$ gives $\widehat{\boldsymbol{\beta}}_{\text{init}} - \boldsymbol{\beta}_{\tau,0} = O_p(n^{-1/2}) + O(J_n^{-s_m}) = o_p(n^{-1/4})$ when $J_n \gg n^{1/(4s_m)}$, i.e. the preliminary is $o_p(n^{-1/4})$ (not root- n when $1 < s_m \leq 2$), and supplement I's one-step debiases b_K to first order.

Remark 3 (Estimated weights). For a cross-fitted $\widehat{\omega}^{(-k)}$ the Gram perturbation is $\|\widehat{H}_n(\widehat{\omega}) - \widehat{H}_n(\omega_0)\|_{\text{op}} \leq \|\widehat{\omega} - \omega_0\|_\infty \cdot \mathbb{P}_n[\Delta^T \sum_\ell f_\ell(0) \|\mathbf{Z}_{i\ell}\|^2]$, so Gram consistency is preserved when $\|\widehat{\omega} - \omega_0\|_\infty = o_p(1)$. The profiled-score weight perturbation splits into an empirical part, controlled conditionally by cross-fitting, and a population part that is 0 at the exact conditional quantile and otherwise $O_p(r_\omega(r_m + r_\beta))$ by the conditional-CDF expansion (supplement I, Lemma 5). For CC, $r_\omega = 0$; for IPCW, censoring positivity converts $\|\widehat{G}_C - G_{C,0}\|_\infty$ into the same-order bound for $\|\widehat{G}_C^{-1} - G_{C,0}^{-1}\|_\infty$.

Remark 4 (Finite-moment variant). If only a q th cluster-envelope moment ($q > 4$) is assumed instead of (A6), truncate the envelope at $b_n = n^{1/q}(\log n)^2$: on $\{M_i(1 + \|\mathbf{X}_i\|)^2 \leq b_n\}$ apply Lemmas 9 and 11 with the truncated summands, and bound the discarded tail by Markov, $\mathbb{P}(\exists i : M_i(1 + \|\mathbf{X}_i\|)^2 > b_n) \leq n b_n^{-q} \mathbb{E}[\dots] \rightarrow 0$. The conclusions hold with $\delta_{H,n}$ and the maximal-inequality rate multiplied by $b_n/\sqrt{K_n}$, which only tightens the sieve-growth condition by a logarithmic-polynomial factor.

S1.0.15 The admissible sieve window

For normalized tensor B-splines, $K_n = J_n^2$ and $\zeta_n \asymp \sqrt{K_n} = J_n$.

Proposition 4 (Non-vacuous window for $s > 1$). *Under Condition 1, the requirement $\delta_{H,n} r_n = o(n^{-1/2})$ of Lemma 13 (with $r_n \asymp \sqrt{K_n/n}$) is implied by*

$$\frac{K_n^2 \log K_n}{n} \rightarrow 0 \quad \Longleftrightarrow \quad \frac{J_n^4 \log J_n}{n} \rightarrow 0,$$

and, together with $J_n^{-s} = o(n^{-1/4})$, yields the non-empty window

$$n^{1/(4s)} \ll J_n \ll \left(\frac{n}{\log n}\right)^{1/4},$$

non-empty precisely for $s > 1$.

Proof. $\delta_{H,n} r_n = (\sqrt{K_n \log K_n/n} + K_n \log K_n/n) \sqrt{K_n/n}$. The leading term is $\sqrt{K_n^2 \log K_n/n^2} = (K_n/n) \sqrt{\log K_n}$, which is $o(n^{-1/2})$ iff $K_n \sqrt{\log K_n} = o(\sqrt{n})$, i.e. $K_n^2 \log K_n/n \rightarrow 0$; the second term is then lower order. With $K_n = J_n^2$ this is $J_n^4 \log J_n/n \rightarrow 0$, i.e. $J_n \ll (n/\log n)^{1/4}$. The bias side $J_n^{-s} = o(n^{-1/4})$ requires $J_n \gg n^{1/(4s)}$. The window is non-empty iff $1/(4s) < 1/4$, i.e. $s > 1$. $\square$

Remark 5 (Local support is essential, not cosmetic). If one ignores local support and bounds each normalized row crudely by $\|B_n\| = O(\sqrt{K_n})$, a full-matrix Bernstein bound gives $\|\hat{H}_n - H_n\|_{\text{op}} = O_p(K_n \sqrt{\log K_n/n})$, forcing $K_n^3 \log K_n/n \rightarrow 0$, i.e. $J_n = o(n^{1/6})$ and $s > 3/2$, which is *incompatible* with the $s > 1$ window. Exploiting the bounded-overlap structure (A4)— $O(K_n)$ nonzero pairs and operator-norm = max row sum—is what reduces the Gram rate to $\sqrt{K_n \log K_n/n}$ and rescues $s > 1$.

S1.0.16 Summary

Lemmas 9–13 and Proposition 4 establish, for the oracle weight ω_0 and (under Remark 3) the cross-fitted estimated weight, the parametric Bahadur representation that supplies conditions C5–C7 of supplement I’s Theorem 1. The only inputs are the regularity Condition 1; the formerly “open” clustered maximal inequality and Gram concentration are proved here, the decisive step being the local-support Gram bound of Lemma 9.

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S2 Algorithms and implementation

Sieve basis. The baseline surface $m_\tau(u, t)$ uses a tensor-product cubic B-spline basis $\{B_k(u, t)\}$ over the two time scales, a partition of unity (so no separate intercept enters the design). The implementation uses this partition-of-unity basis; the asymptotic analysis (Section S1) uses the L_2 -normalized basis $\tilde{B}_k = \sqrt{K} B_k$, under which the population Gram has eigenvalues bounded away from 0 and ∞ ; the two differ by the fixed scaling $\sqrt{K}$, which cancels in the estimator.

Complexity selection (A-CV). The number of interior knots / spline degrees of freedom and any penalty λ_n are chosen by subject-level K -fold cross-validation minimizing the held-out check loss (the “-CV” suffix). A fixed richer-sieve fit (“A-Rich”, $K+2$) is reported as a sensitivity comparison, not a theory-guided undersmoothing rule. The reported pilot uses a regression-spline sieve (fixed knots, $\lambda_n = 0$).

Sparsity and projection (B-OS-CF). The conditional density at the quantile, $f_\tau(0)$, is estimated by the Hendricks–Koenker quantile-spacing sparsity $\hat{f}_\tau(0) = \{(\hat{q}_{\tau+a_n} - \hat{q}_{\tau-a_n})/(2a_n)\}^{-1}$, rearranged for positivity and truncated to $[f_{\min}, f_{\max}]$. The density-weighted projection $\hat{g}_{\tau,\omega}$ solves a weighted spline least-squares regression of $\mathbf{X}$ on $B_g(u, t)$ with weights $a_\omega \hat{f}_\tau(0)$. Cross-fitting over subject folds is part of the estimator definition (Section S1.0.4 of the main paper).

Bootstrap and fold discipline. Inference for the A-CV point estimator uses a full-pipeline subject-level bootstrap: subjects (not observations) are resampled, preserving within-subject clustering, and the entire pipeline (complexity selection and, for B-OS-CF, nuisance fitting) is refit on each resample. Cross-validation folds are formed by subject id, so no subject’s visits appear in both training and held-out folds; a controlled same-dataset comparison (Table S19) quantifies the effect of violating this fold discipline.

Crossing. Predicted quantile curves are made monotone in τ by rearrangement; this is a pointwise post-processing step that does not alter the raw $\hat{\beta}_\tau$ used for inference.

Reproducibility and computational audit. A computational-failure audit (Table S25) records the disposition of every replicate (QR-solver status, degenerate folds, support-mask exclusions) so that reported summaries are not silently conditioned on “clean” replicates. Code is available at the companion repository.

S3 Additional simulations

This section gives the full designs and complete tables for the studies summarized in Section 5 of the main paper. Unless noted, R denotes the number of Monte-Carlo replicates and coverage is for the 95% subject-bootstrap interval.

S3.1 Estimator family and finite-sample inference

S3.2 Censoring weights and orthogonality

The complete-event (CC), IPCW, and oracle-weight (ORC) estimators, the estimated-vs-oracle weight-gap rate, and hard-censoring behavior.

S3.3 Surface recovery and density-weighted centering

S3.4 Sieve growth, tuning, and cross-validation behavior

S3.5 Fold discipline and randomization

S3.6 Non-Gaussian errors and crossing

S3.7 Onset-distribution robustness (non-proportional hazards / AFT) and pilot

S3.8 Computational reproducibility audit

S3.9 Comparison with the conditional-mean estimator of Sun et al.

Full numbers for the five-regime comparison of Section 5.1. Table S26 reports distributional-effect recovery and scalar-slope variability (with the oracle- α decomposition discussed in the main text); Table S27 the mean-contrast IMSE for Sun-basic, the stratified Sun-S extension, and the two quantile reconstructions Q-LN and Q-INT, with paired Monte-Carlo standard errors; Table S28 the conditional-mean surface and Sun bandwidth sensitivity; and Table S29 the heavy-tail (DGP5) robustness diagnostics.

Table S1: Headline estimator comparison on a *single* paired experiment ($R = 1000$; bootstrap coverage over a 500-replicate subset, $B = 500$ full-refit resamples; MC SE ≈ 0.010). Four estimators on the same replicates: complete-event (CC, $\omega = 1$) and IPCW ($\omega = 1/\hat{G}_C$), each as the joint-sieve A-CV point estimator and the cross-fitted one-step **B-OS-CF** ($K = 5$ folds). All coverages are full-refit subject bootstrap.

Sc.	n	τ	$\beta_{\tau,1}$	bias				bootstrap coverage			
				CC-ACV	IP-ACV	CC-CF	IP-CF	CC-ACV	IP-ACV	CC-CF	IP-CF
1	500	0.25	0.500	-0.002	-0.002	0.000	-0.000	0.960	0.958	0.970	0.968
1	500	0.50	0.500	-0.000	0.001	0.002	0.002	0.968	0.962	0.984	0.968
1	500	0.75	0.500	-0.002	-0.003	-0.008	-0.009	0.950	0.954	0.966	0.962
1	1000	0.25	0.500	0.004	0.004	0.004	0.005	0.938	0.942	0.960	0.960
1	1000	0.50	0.500	-0.001	-0.001	-0.001	-0.001	0.946	0.948	0.954	0.960
1	1000	0.75	0.500	0.002	0.002	0.001	0.003	0.958	0.946	0.958	0.972
1	2000	0.25	0.500	0.000	-0.000	0.001	0.001	0.956	0.952	0.970	0.966
1	2000	0.50	0.500	-0.000	-0.000	0.000	-0.000	0.966	0.956	0.966	0.978
1	2000	0.75	0.500	0.002	0.002	0.002	0.003	0.942	0.944	0.972	0.966
2	500	0.25	0.230	0.006	0.007	0.004	0.002	0.918	0.928	0.954	0.952
2	500	0.50	0.500	-0.002	-0.003	-0.004	-0.004	0.950	0.954	0.948	0.948
2	500	0.75	0.770	-0.005	-0.005	-0.002	-0.003	0.954	0.958	0.954	0.958
2	1000	0.25	0.230	0.001	0.003	-0.001	-0.001	0.940	0.932	0.970	0.974
2	1000	0.50	0.500	0.002	0.003	0.002	0.003	0.946	0.938	0.972	0.962
2	1000	0.75	0.770	-0.005	-0.007	-0.004	-0.005	0.946	0.946	0.958	0.970
2	2000	0.25	0.230	-0.001	-0.001	-0.002	-0.001	0.938	0.940	0.960	0.956
2	2000	0.50	0.500	-0.000	0.000	-0.001	-0.000	0.944	0.954	0.964	0.956
2	2000	0.75	0.770	0.002	0.002	0.003	0.004	0.948	0.950	0.980	0.986

S4 Additional application details

Synthetic cohort. The illustrative analysis of Section 6 uses a synthetic cohort generated to match the ADNI sampling design: covariates are a binary *APOE4* carrier indicator and sex; onset (amyloid positivity) is Cox–Weibull; competing death is Weibull; entry is staggered with left truncation; loss to follow-up is independent given $(A, \mathbf{X})$; visits are noninformative. The log-marker is generated with a covariate effect that varies across quantile levels so that the carrier effect $\beta_{\tau,1}$ is non-flat in τ , illustrating a distributional (not pure-location) effect. *These are not real ADNI data and the figures are for illustration of the deliverables only; the substantive ADNI analysis is in preparation.*

Estimable region and masks. Quantile surfaces are reported on the interior region where the visit occupation density is bounded away from zero; grid cells with fewer than a minimum number of nearby complete-event visits are masked.

Sensitivity. For the real-data analysis we will report sensitivity to (i) the weighting (CC default vs. IPCW design-standardization), (ii) the inferential route (A-CV with subject bootstrap vs. B-OS-CF), and (iii) crossing (rearrangement of predicted curves), following the same protocol validated in Section S3.

Table S2: Reporting-completeness supplement to Tables S1/S3: the Monte-Carlo ESD and mean bootstrap SE ($\overline{\text{SE}}$) for all four estimators, reported *directly* so neither need be backed out of SER/RMSE ($R = 1000$, $R_{\text{cov}} = 500$). Within each pipeline B-OS-CF has a larger ESD than A-CV. The B-OS-CF–A-CV *difference* is dominated in *magnitude* by the orthogonal correction, with a smaller sample-splitting contribution (Table S8: $|c_{\text{os}}|/\text{ESD} \approx 0.28\text{--}0.40$ vs $|c_{\text{split}}|/\text{ESD} \approx 0.07\text{--}0.13$); we do *not* decompose the variance of the ESD increase itself (that would require the covariance terms with $\hat{\beta}_{\text{ACV}}$). that $\overline{\text{SE}} \gtrsim \text{ESD}$ for B-OS-CF is consistent with bootstrap-SE overestimation being an important contributor to its mild over-coverage. Bias and coverage are in Table S1; SER and RMSE in Table S3.

Sc.	n	τ	CC A-CV		CC B-OS-CF		IP A-CV		IP B-OS-CF	
			ESD	$\overline{\text{SE}}$	ESD	$\overline{\text{SE}}$	ESD	$\overline{\text{SE}}$	ESD	$\overline{\text{SE}}$
1	500	0.25	0.110	0.113	0.123	0.129	0.115	0.117	0.131	0.135
1	500	0.50	0.099	0.108	0.106	0.121	0.106	0.112	0.117	0.126
1	500	0.75	0.108	0.113	0.117	0.129	0.112	0.117	0.124	0.134
1	1000	0.25	0.078	0.079	0.087	0.094	0.083	0.083	0.092	0.100
1	1000	0.50	0.073	0.075	0.082	0.088	0.077	0.079	0.086	0.093
1	1000	0.75	0.079	0.080	0.089	0.095	0.083	0.083	0.095	0.100
1	2000	0.25	0.055	0.056	0.059	0.067	0.059	0.059	0.063	0.071
1	2000	0.50	0.049	0.053	0.054	0.061	0.052	0.055	0.058	0.065
1	2000	0.75	0.055	0.056	0.059	0.065	0.058	0.058	0.065	0.070
2	500	0.25	0.144	0.145	0.160	0.167	0.152	0.151	0.168	0.175
2	500	0.50	0.134	0.138	0.150	0.157	0.140	0.144	0.157	0.164
2	500	0.75	0.142	0.146	0.160	0.169	0.148	0.152	0.165	0.176
2	1000	0.25	0.104	0.102	0.113	0.121	0.110	0.107	0.120	0.129
2	1000	0.50	0.094	0.096	0.105	0.112	0.099	0.101	0.111	0.119
2	1000	0.75	0.104	0.102	0.114	0.121	0.109	0.107	0.122	0.129
2	2000	0.25	0.073	0.072	0.078	0.084	0.077	0.075	0.084	0.090
2	2000	0.50	0.068	0.068	0.073	0.078	0.072	0.071	0.078	0.084
2	2000	0.75	0.071	0.072	0.075	0.084	0.074	0.076	0.080	0.090

Table S3: Supplement to Table S1: $\text{SER} = \overline{\text{SE}}_{\text{boot}}/\text{ESD}$ and RMSE for *all four* estimators ($R = 1000$, $R_{\text{cov}} = 500$). Since bias is negligible in these cells, $\text{ESD} \approx \text{RMSE}$ and $\overline{\text{SE}} = \text{SER} \times \text{ESD}$; the underlying ESD and $\overline{\text{SE}}$ are reported directly in Table S2. *Most* coverage dips coincide with a SER slightly below 1 (a mildly small bootstrap SE), *not* bias; the exception is the single lowest-coverage cell (Sc. 2, $n=500$, $\tau=0.25$: cov 0.918), which has $\text{SER} \approx 1.00$ *and* negligible bias (0.006), so its shortfall is neither mean SE nor bias. The studentization diagnostic (Table S11) shows $Z = (\hat{\beta} - \beta_0)/\widehat{\text{SE}}$ is near-Gaussian in its central moments (skew -0.01 , excess kurtosis 0.02) with normal = percentile coverage, so the shortfall is not explained by mean SE, central skew/kurtosis, or a simple SE–error association; it is *consistent with* a mild small-sample upper-tail deviation and shrinks with n . B-OS-CF carries a modestly larger RMSE, generally around 5–15% (the variance cost of the explicit orthogonal adjustment), and—after the fold-leakage fix—a $\text{SER} \geq 1$, mildly to *moderately* conservative in several cells (coverage up to ≈ 0.986).

Sc.	n	τ	A-CV cc		A-CV ip		B-OS-CF cc		B-OS-CF ip	
			SER	RMSE	SER	RMSE	SER	RMSE	SER	RMSE
1	500	0.25	1.03	0.110	1.02	0.115	1.05	0.123	1.03	0.131
1	500	0.50	1.08	0.099	1.06	0.106	1.14	0.106	1.08	0.117
1	500	0.75	1.05	0.108	1.05	0.112	1.10	0.117	1.09	0.124
1	1000	0.25	1.01	0.078	1.00	0.083	1.09	0.087	1.09	0.092
1	1000	0.50	1.03	0.073	1.02	0.077	1.07	0.082	1.08	0.086
1	1000	0.75	1.01	0.079	1.00	0.083	1.06	0.089	1.05	0.095
1	2000	0.25	1.01	0.055	1.00	0.059	1.12	0.059	1.13	0.063
1	2000	0.50	1.07	0.049	1.07	0.052	1.13	0.054	1.13	0.058
1	2000	0.75	1.00	0.055	1.00	0.058	1.10	0.059	1.08	0.065
2	500	0.25	1.00	0.144	0.99	0.152	1.04	0.160	1.04	0.168
2	500	0.50	1.03	0.134	1.02	0.140	1.05	0.150	1.05	0.157
2	500	0.75	1.03	0.142	1.03	0.148	1.05	0.160	1.07	0.165
2	1000	0.25	0.98	0.104	0.97	0.110	1.08	0.113	1.08	0.120
2	1000	0.50	1.02	0.094	1.01	0.099	1.07	0.105	1.07	0.111
2	1000	0.75	0.98	0.104	0.98	0.110	1.06	0.114	1.06	0.122
2	2000	0.25	0.98	0.073	0.98	0.077	1.08	0.078	1.07	0.084
2	2000	0.50	0.99	0.068	0.99	0.072	1.07	0.073	1.07	0.078
2	2000	0.75	1.02	0.071	1.02	0.074	1.13	0.075	1.13	0.080

Table S4: Cross-fitted one-step **B-OS-CF** ($K = 5$), CC and IPCW members. *corr* is the per-cell *mean* signed one-step correction $\tilde{\beta}_{\text{CF}} - \hat{\beta}_{\text{ACV}}$; *std* is the mean of the replicate-wise standardized corrections $\frac{1}{R} \sum_r \text{corr}_r / \widehat{\text{SE}}_r$ (so *corr* and *std* can differ in sign). Both signed summaries are small ($|\text{corr}| \leq 0.007$, $|\text{std}| \leq 0.05$ in every cell), consistent with the A-CV first-order conditions already nearly annihilating the orthogonal score at $\lambda = 0$. *Caveat*: a signed mean near zero can mask per-replicate cancellation; the per-replicate *absolute* correction and its split/one-step decomposition are in Table S8.

Sc.	n	τ	corr_{CC}	$\text{corr}_{\text{IPCW}}$	std_{CC}	std_{IPCW}
1	500	0.25	0.002	0.001	0.025	0.010
1	500	0.50	0.002	0.002	0.019	0.003
1	500	0.75	-0.006	-0.005	-0.047	-0.044
1	1000	0.25	-0.000	0.001	0.006	0.007
1	1000	0.50	0.000	-0.001	0.009	0.005
1	1000	0.75	-0.001	0.001	-0.019	-0.004
1	2000	0.25	0.001	0.001	0.021	0.025
1	2000	0.50	0.000	-0.000	0.008	-0.001
1	2000	0.75	-0.000	0.000	-0.015	0.004
2	500	0.25	-0.003	-0.005	-0.024	-0.032
2	500	0.50	-0.002	-0.001	-0.023	-0.004
2	500	0.75	0.003	0.002	0.013	-0.008
2	1000	0.25	-0.003	-0.004	-0.006	-0.044
2	1000	0.50	0.000	0.000	0.019	0.011
2	1000	0.75	0.001	0.001	0.002	0.009
2	2000	0.25	-0.000	0.000	0.003	0.008
2	2000	0.50	-0.000	-0.000	-0.009	-0.008
2	2000	0.75	0.001	0.001	0.015	0.012

Table S5: **Paired** CC–A-CV vs. IPCW–A-CV on the same replicates and bootstrap draws ($R = 1000$; coverage over 500; $B = 500$). $\Delta\text{cov} = \text{cov}_{\text{CC}} - \text{cov}_{\text{IPCW}}$; McNemar (b, c) are the discordant counts (CC covers & IPCW not; IPCW covers & CC not) with the two-sided *exact* (binomial) McNemar p ; ESD ratio is CC/IPCW. Of the 18 cells, *one* (Sc. 1, $n=1000$, $\tau=0.75$: $b=6, c=0$) gives an unadjusted exact McNemar $p = 0.031$; **no cell remains significant after a Holm/Bonferroni multiplicity correction** (the next smallest is $p = 0.07$). The absolute coverage differences are $|\Delta\text{cov}| \leq 0.012$ with no systematic direction. At $R_{\text{cov}} = 500$ this does not formally establish equivalence at a 0.02 margin (a paired TOST would, and is left to a larger run) but is consistent with it. Separately, the CC ESD is smaller in every cell—an *observed* efficiency improvement under these DGPs (not a claim of provably greater accuracy, which a non-significant test cannot support).

Sc.	n	τ	cov_{CC}	cov_{IPCW}	Δcov	b	c	p	$\frac{\text{ESD}_{\text{CC}}}{\text{ESD}_{\text{IPCW}}}$
1	500	0.25	0.960	0.958	0.002	3	2	1.00	0.955
1	500	0.50	0.968	0.962	0.006	6	3	0.51	0.936
1	500	0.75	0.950	0.954	-0.004	2	4	0.69	0.959
1	1000	0.25	0.938	0.942	-0.004	3	5	0.73	0.942
1	1000	0.50	0.946	0.948	-0.002	3	4	1.00	0.950
1	1000	0.75	0.958	0.946	0.012	7	1	0.07	0.949
1	2000	0.25	0.956	0.952	0.004	4	2	0.69	0.942
1	2000	0.50	0.966	0.956	0.010	6	1	0.13	0.953
1	2000	0.75	0.942	0.944	-0.002	7	8	1.00	0.956
2	500	0.25	0.918	0.928	-0.010	3	8	0.23	0.951
2	500	0.50	0.950	0.954	-0.004	3	5	0.73	0.954
2	500	0.75	0.954	0.958	-0.004	2	4	0.69	0.958
2	1000	0.25	0.940	0.932	0.008	9	5	0.42	0.947
2	1000	0.50	0.946	0.938	0.008	5	1	0.22	0.945
2	1000	0.75	0.946	0.946	0.000	4	4	1.00	0.948
2	2000	0.25	0.938	0.940	-0.002	2	3	1.00	0.949
2	2000	0.50	0.944	0.954	-0.010	1	6	0.13	0.951
2	2000	0.75	0.948	0.950	-0.002	3	4	1.00	0.950

Table S6: Identification/efficiency and the $\chi^C = 0$ diagnostic (A-CV $\beta_{\tau,1}$, $\tau = 0.5$, $R = 400$; base-censoring regime). CC = unweighted complete-event; IPCW = estimated $\hat{G}_C$; ORC = oracle $G_{C,0}$. All three are unbiased; $\text{ESD}_{\text{IPCW}} \approx \text{ESD}_{\text{ORC}}$ (no detectable additional variance from estimating G_C) and $\geq \text{ESD}_{\text{CC}}$. The gap $\text{sd}(\hat{\beta}_{\text{IPCW}} - \hat{\beta}_{\text{ORC}})/\text{ESD}_{\text{IPCW}}$ shrinks with n (n^{-1} -like, $\ll n^{-1/2}$).

Sc.	n	bias_{CC}	$\text{bias}_{\text{IPCW}}$	bias_{ORC}	ESD_{CC}	ESD_{IPCW}	ESD_{ORC}	gap/ESD
1	500	-0.0033	-0.0041	-0.0034	0.1011	0.1057	0.1044	0.1163
1	1000	0.0012	0.0010	0.0011	0.0728	0.0766	0.0762	0.0842
1	2000	-0.0010	-0.0011	-0.0012	0.0482	0.0501	0.0502	0.0710
1	4000	0.0040	0.0045	0.0044	0.0360	0.0377	0.0377	0.0483
2	500	-0.0049	-0.0039	-0.0041	0.1360	0.1460	0.1455	0.1206
2	1000	0.0020	0.0027	0.0031	0.1007	0.1070	0.1065	0.0783
2	2000	-0.0015	-0.0013	-0.0011	0.0664	0.0702	0.0700	0.0621
2	4000	-0.0043	-0.0048	-0.0048	0.0511	0.0540	0.0539	0.0495

Table S7: Censoring-weight stress test: the $\chi^C = 0$ gap under the base regime vs. a *hard* regime (later/wider entry + heavier, more covariate-dependent loss; Sc. 2, $\tau = 0.5$, $R = 400$). The hard regime lifts censoring to $\approx 43\%$ with markedly heavier IPCW-weight tails (99th pct ≈ 5 , max ≈ 12) and lower weight ESS/ n_T (≈ 0.79 vs. 0.98), i.e. closer to the positivity boundary; the Cox $\hat{G}_C$ never failed. Even so, the estimated-vs-oracle gap/ESD stays small relative to the estimator ESD and decreases overall ($0.20, 0.19, 0.13, 0.13$ at $n = 500, 1000, 2000, 4000$); we do *not* claim a clean n^{-1} rate—the final two sample sizes are nearly flat—only that the ratio stays small and decreases overall, which is *consistent with* (but does not establish) the second-order, $\ll n^{-1/2}$ behaviour. ($\tau = 0.5$ only; tail behaviour under hard weights is not probed.)

regime	n	%cens	ESS/ n_T	$\tilde{w}$	w_{95}	w_{99}	$w_{\max}$	gap/ESD
base	500	0.17	0.98	1.17	1.53	1.76	2.07	0.1206
base	1000	0.17	0.98	1.17	1.53	1.75	2.14	0.0783
base	2000	0.17	0.98	1.17	1.53	1.74	2.19	0.0621
base	4000	0.17	0.98	1.17	1.52	1.73	2.29	0.0495
hard	500	0.43	0.79	1.47	3.35	5.10	7.43	0.1997
hard	1000	0.42	0.79	1.47	3.40	5.16	8.48	0.1925
hard	2000	0.43	0.79	1.47	3.42	5.28	9.66	0.1308
hard	4000	0.43	0.78	1.47	3.42	5.34	11.54	0.1297

Table S8: B-OS-CF one-step correction $\tilde{\beta}_{\text{CF}} - \hat{\beta}_{\text{ACV}}$ and its *decomposition* (CC, Scenario 2, $R = 500$). The signed mean is near zero (cancellation), but the per-replicate *absolute* correction is non-trivial (mean $|c|$ is 0.28–0.40 of the A-CV ESD). Decomposing $\text{corr} = c_{\text{split}} + c_{\text{os}}$ into the sample-splitting part $c_{\text{split}} = \sum_k \frac{n_k}{n} \hat{\beta}_{\text{init}}^{(-k)} - \hat{\beta}_{\text{ACV}}$ and the pure one-step part $c_{\text{os}} = \sum_k \frac{n_k}{n} \hat{S}_k^{-1} \hat{U}_k$: the orthogonal part **dominates** (mean $|c_{\text{os}}|/\text{ESD} \approx 0.28$ – 0.40 vs mean $|c_{\text{split}}|/\text{ESD} \approx 0.07$ – 0.13), so the adjustment is genuinely the orthogonal correction, not cross-fitting/splitting noise. Whether it reduces *bias* is separate, addressed in the tail regime of Table S13.

n	τ	mean $ c $	$\frac{ c }{\text{ESD}}$	$\frac{ c_{\text{os}} }{\text{ESD}}$	$\frac{ c_{\text{split}} }{\text{ESD}}$	signed mean
500	0.25	0.052	0.37	0.37	0.12	0.0004
500	0.50	0.051	0.37	0.38	0.11	0.0033
500	0.75	0.056	0.40	0.40	0.13	0.0009
1000	0.25	0.033	0.34	0.34	0.10	-0.0022
1000	0.50	0.032	0.35	0.35	0.09	-0.0004
1000	0.75	0.035	0.37	0.37	0.10	0.0020
2000	0.25	0.022	0.32	0.32	0.09	-0.0011
2000	0.50	0.019	0.28	0.28	0.07	0.0011
2000	0.75	0.020	0.28	0.28	0.08	0.0010

Table S9: Surface-heterogeneity DGP $\sigma(u, t, X) = \sigma_0 + \sigma_1 X_1 + \sigma_2(t - u) + \sigma_3(u/t)$, so $m_{\tau,0}(u, t)$ is *non-parallel* across τ (**fitted estimator: CC–A–CV**; $R = 500$, coverage over 300, $B = 300$). “het” is the base design; “stress” adds high truncation, 32% censoring, ~ 1.3 visits/subject. **Both** slopes are reported: bias and bootstrap coverage of $\beta_{\tau,1}$ and the continuous-covariate $\beta_{\tau,2}$. IMSE columns are Monte-Carlo *means*: IMSE_{unm} over the whole grid, and—now shown side by side— $\text{IMSE}_{\text{msk}_v}$ over the *visit*-estimable cells and $\text{IMSE}_{\text{msk}_s}$ over the *distinct-subject*-estimable cells. The two local-support masks: “%n.e. visit” (fewer than 10 pre-onset *visits* within ± 0.5 in (u, t)) and the surface-relevant “%n.e. subj” (fewer than 5 *distinct subjects*); both are replicate-specific and design-only (not strictly comparable across n , and the two thresholds are not information-equivalent—the subject mask is looser by construction). Because visits cluster within subject, the subject mask flags about *half* as many cells as the visit mask (stress: $\approx 21\%$ vs. $\approx 41\%$), yet the two masked IMSEs are nearly identical (columns msk_v vs. msk_s , both ≈ 0.10 under stress), so the two operational masks give *qualitatively similar* conclusions—not a formal mask-invariance claim, since the thresholds are not information-matched (the subject mask is looser) and each mask is replicate-specific (a fixed common-domain IMSE would be needed to compare convergence across n). $\text{ESS}^{\text{ip}}/n_T$ is a *design diagnostic only* (the corresponding IPCW weights; the fitted CC estimator has $w \equiv 1$). β stays broadly covered (finite-sample bias rises somewhat in the smallest stress sample); the low-information boundary surface should be read only on the estimable domain.

set	n	τ	β_1		β_2		IMSE			%n.e.		$\frac{\text{ESS}^{\text{ip}}}{n_T}$
			bias	cov	bias	cov	unm	msk_v	msk_s	visit	subj	
het	500	0.25	-0.001	0.947	0.001	0.933	0.0326	0.0323	0.0324	0.03	0.02	0.98
het	500	0.50	-0.005	0.963	0.003	0.957	0.0285	0.0284	0.0284	0.03	0.02	0.98
het	500	0.75	0.000	0.943	-0.004	0.927	0.0320	0.0316	0.0317	0.03	0.02	0.97
het	1000	0.25	0.005	0.920	-0.002	0.937	0.0159	0.0158	0.0158	0.00	0.00	0.98
het	1000	0.50	0.000	0.957	0.003	0.933	0.0133	0.0133	0.0133	0.00	0.00	0.98
het	1000	0.75	-0.004	0.933	-0.003	0.943	0.0158	0.0158	0.0158	0.00	0.00	0.98
het	2000	0.25	0.003	0.937	0.001	0.927	0.0077	0.0077	0.0077	0.00	0.00	0.98
het	2000	0.50	-0.000	0.933	0.003	0.957	0.0067	0.0067	0.0067	0.00	0.00	0.98
het	2000	0.75	0.003	0.927	-0.000	0.947	0.0074	0.0074	0.0074	0.00	0.00	0.98
stress	500	0.25	0.019	0.947	0.010	0.947	0.1207	0.0956	0.1006	0.41	0.21	0.88
stress	500	0.50	0.002	0.957	-0.004	0.973	0.0976	0.0811	0.0840	0.40	0.21	0.86
stress	500	0.75	-0.026	0.947	-0.007	0.957	0.1370	0.0900	0.0952	0.40	0.21	0.88
stress	1000	0.25	0.013	0.953	-0.003	0.940	0.0522	0.0459	0.0467	0.14	0.07	0.90
stress	1000	0.50	-0.003	0.940	0.001	0.953	0.0422	0.0384	0.0390	0.14	0.07	0.88
stress	1000	0.75	-0.008	0.943	-0.001	0.957	0.0508	0.0463	0.0470	0.14	0.07	0.88
stress	2000	0.25	0.007	0.960	0.002	0.943	0.0231	0.0221	0.0224	0.04	0.02	0.87
stress	2000	0.50	-0.000	0.943	-0.000	0.937	0.0210	0.0200	0.0204	0.04	0.02	0.90
stress	2000	0.75	-0.010	0.930	-0.001	0.940	0.0253	0.0242	0.0246	0.05	0.02	0.91

Table S10: Density-weighted (DW) vs. ordinary (ORD) centering (Scenario 2; **all three τ fit on the same dataset per replicate**, $R = 600$ per n ; IPCW weight, *oracle* f , B-OS-*NCF*; only the centering g differs—Sc. 1 has $g_{\text{DW}}=g_{\text{ORD}}$ and is omitted). DW covers better in *every* cell ($\Delta_{\text{cov}} = +0.01$ to $+0.03$; all discordances favour DW, $c=0$): the paired exact McNemar is significant in **8/9** cells unadjusted, **4/9** after Holm, **3/9** after Bonferroni (we report only these cellwise/multiplicity-adjusted results; a single global p is omitted as hard to audit and unnecessary). **The effect is bread calibration, not a different estimator:** DW and ORD point estimates coincide to $\sim 10^{-4}$, the DW interval is only $\approx 6\%$ longer (length/SE ratio ≈ 1.06) and *nests* the ORD interval in $\geq 99\%$ of (rep, τ). Its proper 95% interval score (IS; lower=better; width + miss penalty) is *statistically indistinguishable* from ORD’s: the per-replicate-clustered mean $\overline{\Delta\text{IS}} = -0.0012$ (95% CI $[-0.0050, 0.0025]$, includes 0), and cellwise $\Delta\text{IS} = \text{IS}_{\text{DW}} - \text{IS}_{\text{ORD}}$ is within ≈ 1.6 MCSE of 0 *except* the smallest cell ($n=500, \tau=0.25$: $+0.011$, 2.4 MCSE, where the wider DW interval is slightly *worse*-scored). **So DW improves coverage through a modestly larger sandwich SE, with no detectable change in the proper interval score** (the cellwise ΔIS are small and not uniformly favourable). The clustered $\overline{\Delta\text{IS}}$ averages ΔIS over the three τ *within* each dataset and treats the independent datasets (across n) as the clusters; the resulting CI targets the *equally-weighted* average ΔIS over the nine evaluated (n, τ) design cells (a simulation-design summary, not a single-distribution parameter). This is an *oracle- f* , non-cross-fitted diagnostic and does not by itself carry to the feasible B-OS-CF. Coverage = plug-in sandwich.

n	τ	coverage		int. length		int. score		ΔIS
		DW	ORD	DW	ORD	DW	ORD	
500	0.25	0.965	0.952	0.568	0.535	0.630	0.619	0.0112
500	0.50	0.953	0.925	0.541	0.509	0.624	0.628	-0.0049
500	0.75	0.950	0.930	0.569	0.535	0.692	0.691	0.0003
1000	0.25	0.950	0.937	0.409	0.386	0.484	0.486	-0.0020
1000	0.50	0.953	0.943	0.389	0.366	0.474	0.473	0.0012
1000	0.75	0.938	0.918	0.409	0.385	0.502	0.507	-0.0046
2000	0.25	0.940	0.923	0.291	0.275	0.363	0.368	-0.0053
2000	0.50	0.948	0.928	0.276	0.261	0.327	0.332	-0.0044
2000	0.75	0.952	0.935	0.291	0.275	0.343	0.346	-0.0022

Table S11: Studentization diagnostic for the headline CC-A-CV estimator (Scenario 2, where coverage dips concentrate; $R = 400$, $B = 250$ per replicate). For the standardized statistic $Z_r = (\hat{\beta}_r - \beta_0)/\widehat{SE}_r$ we report its skewness, excess kurtosis, and 2.5/97.5% quantiles; the normal-CI miss rates by side (tr<: truth below the CI, $Z > 1.96$; tr>: truth above, $Z < -1.96$); the cross-replicate correlation $\rho(|\hat{\beta} - \beta_0|, \widehat{SE})$ (large negative $\Rightarrow$ big errors paired with small SE, which would hide undercoverage); and the coverage of the *normal*-SE, *percentile*, and *basic* bootstrap CIs. **At the lowest-coverage cell** ($n=500$, $\tau=0.25$) Z has near-Gaussian central moments (skew -0.01 , excess kurtosis 0.02) with $\rho \approx 0$, and normal (0.93) and percentile (0.93) CIs agree—so the dip is *not* explained by skew/kurtosis, SE-error dependence, or the choice of CI form. The results *suggest* a finite-sample upper-tail deviation ($z_{.975}=2.34 > 1.96$); the side-specific miss-rate difference (tr< 0.043 vs tr> 0.028 , i.e. 0.015 ± 0.013) is imprecise and not clearly distinguishable from 0, and the deviation shrinks with n . Scenario 1 (omitted) is cleaner. The diagnostic gives *no* evidence that percentile intervals improve on the normal-SE interval (the basic interval is worse); both show a similar mild finite-sample shortfall, *consistent with* a finite-sample upper-tail deviation of the studentized statistic. The $z_{.975}=2.34$ rests on only ~ 10 tail replicates ($R=400$), so read it qualitatively; this is a supporting re-run (cov ≈ 0.93), not a per-replicate decomposition of the headline cell.

n	τ	Z shape		Z quant.	CI miss		$\rho_{e,SE}$	coverage		
		skew	ex.kurt	$z_{.025}/.975$	tr<	tr>		norm	pct	basic
500	0.25	-0.01	0.02	-2.00,2.34	0.043	0.028	-0.02	0.930	0.932	0.902
500	0.50	-0.09	0.18	-1.90,1.94	0.022	0.022	0.10	0.955	0.965	0.938
500	0.75	-0.09	0.24	-1.96,1.70	0.020	0.028	0.00	0.953	0.965	0.927
1000	0.25	0.14	0.30	-1.97,2.24	0.040	0.030	0.04	0.930	0.935	0.915
1000	0.50	-0.05	-0.17	-1.95,1.95	0.025	0.025	0.08	0.950	0.950	0.917
1000	0.75	-0.04	0.31	-1.97,2.05	0.028	0.028	-0.05	0.945	0.943	0.927
2000	0.25	-0.03	0.25	-1.89,2.15	0.040	0.025	0.08	0.935	0.943	0.925
2000	0.50	-0.36	0.42	-2.18,1.99	0.030	0.030	-0.00	0.940	0.932	0.935
2000	0.75	0.06	-0.09	-1.83,1.82	0.025	0.020	0.04	0.955	0.960	0.932

Table S12: Sieve asymptotic check WITH inference, CC joint-sieve with *deterministic* J (Scenario 2; $R = 400$, coverage over 250, $B = 250$; $\tau \in \{0.1, 0.5, 0.9\}$). $\sqrt{n}$ bias is shown with its Monte-Carlo SE; † marks cells where $|\sqrt{n} \text{ bias}| > 1.96 \text{ MCSE}$ (a *detectable* scaled bias). $\text{SER} = \overline{\text{SE}}_{\text{boot}}/\text{ESD}$ and the mean 95% interval length disentangle the coverage behaviour. **At $n=500$ the *growing-sieve* bootstrap is degenerate** (mean SER 5–8, mean interval length 3–5 vs. an ESD-implied length ≈ 0.6 , *at all three τ including the median*): a few resamples produce enormous $\widehat{\text{SE}}$ that dominate the *mean*, so the near-0.95 coverage is an artifact of rare very-long intervals, not valid inference. We therefore mark these cells ‡ **not interpretable** (n.i.). The distributional audit (Table S21) shows the *median* bootstrap SE is sound (equal to the exact sieve’s) and the blow-up (SE up to 96) is confined to 2/250 datasets (0.8%, with substantial binomial uncertainty); that audit does *not* attribute these events to the measured Gram-rank, local-support, or onset-diversity diagnostics, and the precise numerical trigger remains *unidentified* (it audits $\tau=0.5$; the same $n=500$ growing-sieve regime shows similar blow-ups at $\tau=0.1, 0.9$, but we do not claim an identical mechanism across τ). The growing sieve is a diagnostic only—the *recommended* procedure uses the small fixed sieve, where this does not occur. For $n \geq 2000$ the SER is well-behaved, and there the low-bias/low-coverage cells (e.g. growing $\tau=0.1$, $n=4000$: $\sqrt{n}$ bias 0.52, SER 0.96, cov 0.896) cannot be explained by the estimated bias (b_K) alone—the scaled bias is only ≈ 0.13 ESD—and appear to reflect studentization/SE distortion as well.

sieve	τ	n	J	$\sqrt{n}$ bias	$\sqrt{n}$ ESD	SER	len	cov ^{boot}
exact	0.10	500	4	0.45 [†] $\pm$ 0.18	3.67	1.00	0.645	0.924
exact	0.10	1000	4	0.19 $\pm$ 0.19	3.79	1.00	0.469	0.936
exact	0.10	2000	4	0.05 $\pm$ 0.20	3.92	0.97	0.331	0.936
exact	0.10	4000	4	-0.21 $\pm$ 0.19	3.71	1.01	0.233	0.948
exact	0.50	500	4	-0.27 $\pm$ 0.15	2.96	1.06	0.549	0.936
exact	0.50	1000	4	0.01 $\pm$ 0.15	2.90	1.06	0.380	0.956
exact	0.50	2000	4	-0.05 $\pm$ 0.15	2.94	1.03	0.266	0.960
exact	0.50	4000	4	0.17 $\pm$ 0.15	3.08	0.99	0.189	0.940
exact	0.90	500	4	-0.26 $\pm$ 0.18	3.67	1.02	0.658	0.940
exact	0.90	1000	4	-0.31 $\pm$ 0.18	3.67	1.01	0.460	0.932
exact	0.90	2000	4	0.05 $\pm$ 0.20	3.94	0.95	0.328	0.944
exact	0.90	4000	4	-0.33 $\pm$ 0.18	3.65	1.03	0.234	0.944
grow	0.10	500	7	0.92 [†] $\pm$ 0.18	3.64	5.22	3.328	n.i. [‡]
grow	0.10	1000	7	0.78 [†] $\pm$ 0.18	3.62	1.03	0.462	0.952
grow	0.10	2000	8	1.03 [†] $\pm$ 0.20	3.93	0.96	0.331	0.944
grow	0.10	4000	9	0.52 [†] $\pm$ 0.20	3.94	0.96	0.233	0.896
grow	0.50	500	7	-0.10 $\pm$ 0.15	3.07	6.66	3.585	n.i. [‡]
grow	0.50	1000	7	0.08 $\pm$ 0.15	2.92	1.06	0.382	0.960
grow	0.50	2000	8	-0.07 $\pm$ 0.15	3.08	1.00	0.269	0.944
grow	0.50	4000	9	-0.01 $\pm$ 0.16	3.22	0.95	0.190	0.940
grow	0.90	500	7	-0.70 [†] $\pm$ 0.18	3.67	7.64	4.911	n.i. [‡]
grow	0.90	1000	7	-0.61 [†] $\pm$ 0.20	3.94	0.96	0.467	0.964
grow	0.90	2000	8	-0.48 [†] $\pm$ 0.18	3.54	1.05	0.325	0.968
grow	0.90	4000	9	-0.46 [†] $\pm$ 0.19	3.81	0.98	0.232	0.928

Table S13: Does the one-step reduce the growing-sieve tail *bias*? Deterministic growing sieve, same replicates through A-joint-sieve, cross-fitted **B-OS-CF**, and a richer-sieve A-Rich ($J+2$) sensitivity (CC, Sc. 2; $R = 400$, coverage over 150, $B = 120$ —a *pilot-grade* inference subset, coverage MCSE ≈ 0.018 ; $\sqrt{n}$ bias MCSE ≈ 0.15 – 0.20). At *both* tails B-OS-CF consistently moves the scaled bias toward zero, increasingly with n (e.g. $\tau=0.1$: $0.66 \rightarrow 0.28$, $0.53 \rightarrow 0.19$; $\tau=0.9$: $-0.49 \rightarrow -0.12$ at $n=4000$), whereas the fixed- $(J+2)$ richer sieve *worsens* it; paired Monte-Carlo intervals would be needed to resolve which individual moves are significant. **However**, SER and RMSE show a *bias*-targeted correction only: B-OS-CF’s SER ≈ 0.93 – 0.97 , coverage is not improved over the deterministic A-joint-sieve (*not* the CV-selected A-CV; the comparator here is the fixed growing sieve), and RMSE is essentially unchanged (the bias drop is offset by a small variance rise)—the scaled bias is already small relative to the SE here, so debiasing is a safeguard, not a coverage fix at these n .

τ	n	J	A-joint-sieve				B-OS-CF				A-Rich	
			$\sqrt{n}$ bias	SER	RMSE	cov	$\sqrt{n}$ bias	SER	RMSE	cov	$\sqrt{n}$ bias	cov
0.10	1000	7	1.09	0.98	0.126	0.953	1.01	0.93	0.128	0.913	1.36	0.907
0.10	2000	8	0.66	0.96	0.088	0.973	0.28	0.93	0.089	0.980	0.91	0.960
0.10	4000	9	0.53	0.97	0.062	0.947	0.19	0.96	0.064	0.940	0.68	0.920
0.50	1000	7	0.17	1.03	0.095	0.947	0.16	1.02	0.095	0.940	0.16	0.953
0.50	2000	8	-0.09	0.97	0.070	0.940	-0.15	0.96	0.072	0.953	-0.12	0.933
0.50	4000	9	-0.13	1.00	0.048	0.940	-0.08	0.99	0.050	0.953	-0.14	0.947
0.90	1000	7	-0.88	0.98	0.125	0.967	-0.75	0.95	0.126	0.973	-1.23	0.967
0.90	2000	8	-0.16	1.02	0.083	0.947	0.09	0.95	0.089	0.927	-0.33	0.940
0.90	4000	9	-0.49	1.02	0.059	0.967	-0.12	0.97	0.062	0.960	-0.67	0.967

Table S14: Tuning sensitivity to the sparsity bandwidth $a_n \in \{.03, .05, .08\}$ and projection dimension $df_g \in \{3, 4, 5\}$ (defaults .05, 4), now over the FULL grid $n \in \{500, 1000, 2000\} \times \tau \in \{.1, .5, .9\}$ (Sc. 2, $R = 400$). Per (n, τ) we report the *range* across the 9 combinations of: B-OS-CF $\beta_{\tau,1}$ bias and ESD; the plug-in sandwich SER and its coverage; the worst $\hat{f}$ -cap rate; and the median bread condition number. **Point estimation is robust to tuning everywhere** (bias range ≤ 0.03 , ESD range ≤ 0.03 , even at $n=500$ and the tails). **Inference is not:** at $n=500$ the sandwich SER ranges 0.57–1.03 (coverage 0.69–0.93) across the grid—confirming that the a_n -dependent $\hat{f}$ destabilizes the analytic SE at small n (the bootstrap is used for inference instead). This tuning-sensitivity of the analytic sandwich *persists at the tails even at $n=2000$* (for $\tau=0.1, 0.9$, SER ranges 0.89–1.24, coverage 0.91–0.98 across the grid); only the central quantile is comparatively stable there. Note this varies the *sandwich*, whereas the recommended procedure uses the bootstrap—so it bounds the analytic SE’s tuning-robustness, not the bootstrap’s. $\hat{f}$ -cap rates are $\leq 2.2\%$ and $\hat{S}$ is well-conditioned ($\kappa \leq 7$) throughout.

n	τ	bias / ESD range		SER / cov range		$\hat{f}$ -cap	$\kappa_{\max}$
		bias rng	ESD rng	SER rng	cov rng		
500	0.10	[-0.006,0.015]	[0.176,0.203]	[0.57,1.02]	[0.71,0.93]	0.022	7
500	0.50	[-0.015,0.013]	[0.143,0.153]	[0.68,0.95]	[0.76,0.93]	0.021	6
500	0.90	[-0.019,0.002]	[0.177,0.207]	[0.61,1.03]	[0.69,0.92]	0.022	6
1000	0.10	[-0.009,0.018]	[0.130,0.142]	[0.71,1.17]	[0.79,0.97]	0.009	5
1000	0.50	[-0.009,0.012]	[0.096,0.106]	[0.84,1.03]	[0.89,0.95]	0.008	5
1000	0.90	[-0.010,0.006]	[0.128,0.144]	[0.73,1.19]	[0.80,0.98]	0.008	5
2000	0.10	[-0.010,0.009]	[0.088,0.099]	[0.91,1.24]	[0.93,0.98]	0.003	5
2000	0.50	[-0.005,0.002]	[0.069,0.074]	[0.90,1.01]	[0.92,0.95]	0.003	4
2000	0.90	[-0.009,0.002]	[0.086,0.104]	[0.89,1.21]	[0.91,0.98]	0.003	4

Table S15: Variance-ladder for the plug-in clustered sandwich SER ($= \overline{\text{SE}}/\text{Monte-Carlo ESD}$; $R = 500$). $\hat{f}/f_0$ is the per-cell *median over replicates and rows* of the sparsity ratio. Rungs replace plug-in nuisances by truth one at a time, holding the A-CV point estimator fixed; the g -true rung uses the *finite-sieve* oracle g_J at the point-estimator’s df; fix- J is a *separate* rung that re-selects the point estimator at a deterministic df (normalized by its own ESD). Both the f -true (0.88–1.07) and oracle- (f, g) (0.89–1.06) rungs are near nominal, while the g -true rung *alone* stays low (0.54–0.95): replacing the sparsity $\hat{f}$ nearly closes the gap, replacing g alone does not. Sparsity estimation is thus the *largest identified contributor* in the small- and moderate- n cells, though residual distortion remains and the f -true improvement is not monotone in every large- n cell; the bread nuisances must be replaced *jointly* to be consistent. Row/cluster < 1 confirms clustering is already included (and matters).

Sc.	n	τ	$\hat{f}/f_0$	plug	f -true	g -true	oracle	fix- J	$\frac{\text{row}}{\text{clus}}$
1	500	0.25	1.04	0.70	0.88	0.54	0.89	0.59	0.63
1	500	0.50	1.04	0.77	0.96	0.60	0.97	0.67	0.61
1	500	0.75	1.02	0.75	0.92	0.58	0.93	0.62	0.63
1	1000	0.25	1.02	0.85	0.95	0.74	0.95	0.79	0.62
1	1000	0.50	1.02	0.89	0.98	0.79	0.98	0.82	0.60
1	1000	0.75	1.02	0.87	0.96	0.76	0.96	0.81	0.62
1	2000	0.25	1.00	0.93	0.97	0.88	0.97	0.90	0.60
1	2000	0.50	1.00	1.02	1.07	0.95	1.06	0.99	0.59
1	2000	0.75	1.00	0.94	0.98	0.87	0.98	0.91	0.60
2	500	0.25	1.03	0.78	0.91	0.55	0.92	0.68	0.63
2	500	0.50	1.02	0.80	0.93	0.64	0.95	0.69	0.61
2	500	0.75	1.04	0.83	0.98	0.64	0.99	0.72	0.64
2	1000	0.25	1.01	0.93	0.99	0.82	1.00	0.87	0.61
2	1000	0.50	1.02	0.86	0.93	0.78	0.93	0.82	0.60
2	1000	0.75	1.01	0.89	0.95	0.79	0.95	0.84	0.61
2	2000	0.25	1.00	0.98	1.00	0.92	1.00	0.95	0.60
2	2000	0.50	1.01	0.98	1.01	0.91	1.01	0.96	0.59
2	2000	0.75	1.00	0.95	0.97	0.88	0.96	0.92	0.60

Table S16: Frequency of the CV-selected spline complexity $\hat{J} \in \{3, 4, 5\}$ (the A-CV tuning), by Scenario, n , and τ ($R = 500$ per cell). Here $\hat{J}$ is the per-margin $\text{df} = \text{degree} + (\text{interior knots})$, so $\hat{J}=3$ is a genuine *cubic* with *zero* interior knots (4 B-spline functions/margin, 16 tensor functions), not a low-degree fit; the adaptive choice is the number of interior knots (defined in the reproducibility specification). **The selection concentrates on the grid minimum $\hat{J} = 3$** (81–86%), with $P(\hat{J}=4) \approx 0.10\text{--}0.15$ and $P(\hat{J}=5) \approx 0.02\text{--}0.04$, and is nearly invariant in n and τ . Two consequences: (i) A-CV is here close to a *fixed* $J=3$ estimator, so its “adaptive” component is small and the A-Rich ($J+2$) contrast is effectively $J=5$ vs. $J=3$ rather than a perturbation around an interior optimum; (ii) the selection saturates at the *minimum* complexity available within the cubic-spline family ($\hat{J}=3$ is zero interior knots, which a wider J grid cannot undercut), so whether a lower-degree or parametric surface would be preferred is not determined by this experiment. This tempers the adaptive-tuning rung of Table S15.

Sc.	n	τ	$P(\hat{J}=3)$	$P(\hat{J}=4)$	$P(\hat{J}=5)$	mode
1	500	0.25	0.81	0.15	0.04	3
1	500	0.50	0.84	0.13	0.03	3
1	500	0.75	0.86	0.12	0.03	3
1	1000	0.25	0.85	0.10	0.04	3
1	1000	0.50	0.83	0.13	0.04	3
1	1000	0.75	0.82	0.14	0.04	3
1	2000	0.25	0.83	0.14	0.04	3
1	2000	0.50	0.85	0.12	0.03	3
1	2000	0.75	0.82	0.15	0.03	3
2	500	0.25	0.82	0.14	0.04	3
2	500	0.50	0.85	0.13	0.02	3
2	500	0.75	0.82	0.14	0.04	3
2	1000	0.25	0.83	0.14	0.04	3
2	1000	0.50	0.84	0.12	0.04	3
2	1000	0.75	0.85	0.13	0.02	3
2	2000	0.25	0.84	0.11	0.04	3
2	2000	0.50	0.84	0.13	0.03	3
2	2000	0.75	0.86	0.11	0.03	3

Table S17: Is the data-adaptive A-CV materially different from a *fixed* small sieve? (Scenario 2, CC estimator, same replicates; $R = 400$, $B = 200$.) Since CV selects $\hat{J}=3$ in 80–86% of replicates (Table S16), we put A-CV (CV-selected $\hat{J}$) head-to-head with fixed $J=3$. **CV genuinely prefers $J=3$ —it is not tie-breaking noise:** the normalized CV-loss gaps ($\text{gap}_J = (\text{CV}(J) - \text{CV}(3))/|\text{CV}(3)|$, the relative increase in held-out *complete-event* (CC, $\omega=1$) check loss—the CC member tunes its df by CC-CV, so *no* $\hat{G}_C$ enters the CC pipeline) are positive ($\text{gap}_4 \approx 0.004\text{--}0.029$, $P(\text{gap}_4 > 0) \approx 0.85$), shrinking with n . **And A-CV $\approx$ fixed $J=3$:** the point estimates differ by only $|\Delta| \approx 0.002\text{--}0.007$, with near-identical bias, ESD, and bootstrap coverage. On the *surface* IMSE, however, fixed $J=3$ is *consistently lower*—by $\approx 9\text{--}13\%$ in the displayed cells; the paired $\overline{\Delta\text{IMSE}} = \text{IMSE}_{\text{ACV}} - \text{IMSE}_{J3}$ (last column, MCSE in parentheses) is positive throughout, consistent with occasional overfitting when A-CV selects $J \in \{4, 5\}$. So in these DGPs **CV mostly reproduces the fixed $J=3$ fit and offers little measurable gain on β** —the headline simulation therefore essentially validates a small fixed sieve. We do not claim CV equals the per-replicate oracle complexity, nor that it “protects” against richer fits (the average CV criterion and the majority of replicates favour the smallest candidate, but $\approx 14\text{--}20\%$ select $J \in \{4, 5\}$); only that its preference for $J=3$ is systematic, not tie-breaking noise.

n	τ	$P(\hat{J}=3)$	CV gap		$ \Delta\beta $	coverage		IMSE		ΔIMSE
			$\frac{\text{CV}_4}{\text{CV}_3}$	$\frac{\text{CV}_5}{\text{CV}_3}$		A-CV	$J=3$	A-CV	$J=3$	
500	0.25	0.81	0.026	0.074	0.0064	0.927	0.935	0.0335	0.0295	0.0040(0.0006)
500	0.50	0.86	0.023	0.059	0.0033	0.950	0.953	0.0273	0.0249	0.0024(0.0004)
500	0.75	0.82	0.026	0.065	0.0062	0.950	0.955	0.0334	0.0300	0.0035(0.0006)
1000	0.25	0.82	0.011	0.027	0.0033	0.930	0.932	0.0160	0.0142	0.0019(0.0003)
1000	0.50	0.81	0.009	0.020	0.0027	0.945	0.943	0.0138	0.0120	0.0018(0.0002)
1000	0.75	0.81	0.012	0.025	0.0032	0.943	0.943	0.0153	0.0138	0.0015(0.0002)
2000	0.25	0.83	0.005	0.010	0.0018	0.938	0.935	0.0078	0.0070	0.0008(0.0001)
2000	0.50	0.83	0.004	0.009	0.0013	0.940	0.943	0.0063	0.0056	0.0007(0.0001)
2000	0.75	0.83	0.005	0.011	0.0018	0.955	0.958	0.0077	0.0069	0.0008(0.0001)

Table S18: Fold-randomization sensitivity of the CC **B-OS-CF** bootstrap (Scenario 2; $R = 250$, $B = 120$, $M = 8$ split draws). The point estimator is the single *realized* fold split (as in the headline) or the *average* over M splits; the bootstrap SE either redraws folds each resample (headline default) or keeps each original subject’s realized fold (*fixed*). SER is relative to the matching point estimator’s ESD. **Redrawing folds inflates the SER by only ≈ 0.01 – 0.015 over fixing them** ($\Delta\text{SER} = \text{SER}_{\text{redr}} - \text{SER}_{\text{fix}}$; paired MCSE in parentheses = the sd of the per-replicate $\widehat{\text{SE}}_{\text{redr}} - \widehat{\text{SE}}_{\text{fix}}$ divided by the common ESD and $\sqrt{R}$, treating ESD as fixed); what matters is the *effect size*—fold redrawing adds only ≈ 0.01 – 0.015 to the SER, a small contribution—so the bulk of the mild B-OS-CF conservativeness is the duplicate-ID fold-leakage correction (Table S19), not fold randomization. The headline bootstrap *redraws* folds each resample and so marginalizes over fold randomness—this is the recommended inference; the fixed-fold and repeated-split rows are *sensitivity checks*. Fixed folds are marginally tighter and empirically well calibrated here (coverage 0.948–0.968); averaging the point over splits lowers its ESD but, with the redrawn SE, *over-covers* (≤ 0.98). Per-cell coverage MCSE ≈ 0.014 ($R=250$), so cell-level coverage gaps should not be over-read.

n	τ	ESD		SER		ΔSER	coverage		
		real	avg	fix	redr		real fix	real redr	avg redr
500	0.25	0.160	0.147	1.03	1.04	0.012(0.005)	0.944	0.952	0.960
500	0.50	0.153	0.137	1.02	1.03	0.013(0.005)	0.956	0.968	0.980
500	0.75	0.157	0.144	1.05	1.07	0.014(0.005)	0.948	0.944	0.968
1000	0.50	0.109	0.101	1.00	1.01	0.013(0.005)	0.968	0.964	0.976
2000	0.50	0.073	0.068	1.05	1.07	0.016(0.005)	0.948	0.948	0.952

Table S19: **Controlled implementation comparison** of the duplicate-ID fold-leakage correction (CC B-OS-CF, Scenario 2; $R = 250$, $B = 150$). On the *same* datasets and resamples the bootstrap SE is computed two ways: *leaked* (the previous implementation—fold by the resampled id, so duplicate copies of a subject split across folds and one trains a nuisance the other is scored against) vs. *clean* (fold by orig_id, duplicates kept together). The last column is the *paired* ratio $\widehat{SE}_{\text{clean}}/\widehat{SE}_{\text{leaked}}$ computed per replicate then averaged (MCSE in parentheses = sd of the per-replicate ratio $/\sqrt{R}$). **The leak systematically tightens the bootstrap SE by 6–11%:** $SER_{\text{leaked}} < 1$ in nearly every cell with coverage dipping to ≈ 0.93 , whereas enforcing fold discipline restores $SER_{\text{clean}} \geq 1$ and near-nominal coverage (0.94–0.98). This attributes most of the mild B-OS-CF conservativeness of Table S1 to the correction; combined with the ≤ 0.015 fold-*randomization* contribution of Table S18, it explains the bulk of the shift from underestimation to mild conservativeness, though residual overcoverage in several cells (clean SER up to 1.17) is *not* uniquely attributable to it. Per-cell coverage MCSE ≈ 0.014 ($R=250$), so cell-level coverage gaps (≈ 0.016 – 0.020) should not be over-read individually.

n	τ	SER		$\frac{\text{clean}}{\text{leaked}}$	coverage	
		leaked	clean		leaked	clean
500	0.25	0.974	1.044	1.076(0.004)	0.932	0.952
500	0.50	0.976	1.036	1.063(0.004)	0.940	0.968
500	0.75	0.999	1.071	1.074(0.005)	0.928	0.948
1000	0.25	0.977	1.075	1.102(0.004)	0.956	0.972
1000	0.50	0.931	1.021	1.099(0.004)	0.944	0.972
1000	0.75	1.033	1.133	1.098(0.004)	0.936	0.952
2000	0.25	0.956	1.057	1.108(0.005)	0.932	0.948
2000	0.50	0.978	1.068	1.093(0.004)	0.932	0.940
2000	0.75	1.065	1.171	1.100(0.004)	0.976	0.984

Table S20: Distributional robustness on the *heteroscedastic* Scenario 2 DGP ($\sigma_1 = 0.4$), so the truth is $\beta_{\tau,1}^{\text{true}} = \beta_{01} + \sigma_1 q_{\tau}(F_G)$ and shifts with the error law F_G (reported as β_1^{tr} for independent verification). The marker error is a standardized non-normal law via a Gaussian copula (latent-Gaussian dependence preserved; raw within-subject correlation not held fixed): “t5” is standardized Student- t_5 , “skew” a standardized right-skew lognormal. $R = 500$, coverage over 300 ($B = 300$; MC SE ≈ 0.013). Point estimation stays approximately unbiased for both slopes; bootstrap coverage is generally satisfactory (0.91–0.98), with moderate under-coverage (min ≈ 0.91) in a few cells.

err	n	τ	β_1^{tr}	bias_{β_1}	ESD_{β_1}	cov_{β_1}	bias_{β_2}	cov_{β_2}
skew	500	0.25	0.222	0.004	0.084	0.960	-0.001	0.963
skew	500	0.50	0.412	-0.004	0.110	0.970	0.002	0.950
skew	500	0.75	0.677	-0.014	0.165	0.947	-0.001	0.910
skew	1000	0.25	0.222	0.002	0.063	0.913	0.001	0.957
skew	1000	0.50	0.412	-0.001	0.076	0.980	-0.001	0.930
skew	1000	0.75	0.677	-0.004	0.118	0.957	0.002	0.943
skew	2000	0.25	0.222	0.001	0.040	0.977	0.001	0.957
skew	2000	0.50	0.412	-0.003	0.057	0.923	-0.001	0.953
skew	2000	0.75	0.677	-0.006	0.083	0.940	-0.001	0.963
t5	500	0.25	0.275	0.007	0.131	0.953	0.001	0.950
t5	500	0.50	0.500	-0.002	0.112	0.947	-0.002	0.967
t5	500	0.75	0.725	-0.004	0.118	0.960	-0.000	0.957
t5	1000	0.25	0.275	0.008	0.090	0.930	0.001	0.950
t5	1000	0.50	0.500	0.005	0.077	0.950	0.004	0.943
t5	1000	0.75	0.725	0.005	0.086	0.960	0.000	0.960
t5	2000	0.25	0.275	0.005	0.066	0.947	0.000	0.957
t5	2000	0.50	0.500	0.001	0.058	0.927	0.001	0.943
t5	2000	0.75	0.725	-0.003	0.062	0.940	0.001	0.937

Table S21: Distributional audit of the deterministic-sieve subject bootstrap at $\tau=0.5$ (same DGP/seed as Table S12; $R = 250$, $B = 150$), with a *scale-invariant* numerical-stability diagnostic. **The median bootstrap SE equals the exact sieve’s at every n** ; only the $n=500$ growing sieve blows up— $\text{SE}_{\max} = 96$, and the $\text{SE} > 3 \times \text{med}$ fraction is 0.8% (2/250 datasets). To test *why*, we replace the raw condition number (which is huge even where the SE is stable) with scale-invariant degeneracy of the standardized design Gram of $[X_1, X_2, \text{surface basis}]$: the *effective rank* $r_{\text{eff}} = (\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ (a participation ratio over the Gram eigenvalues, *not* a numerical rank). Near-zero-support surface columns are removed by a deterministic pre-fit screen; *all* columns surviving that screen are kept in the QR fit (no further numerical-rank truncation, and QR never fails). N_{low} = the count of surface columns carrying < 3 rows of mass. The r_{eff} denominator is the design width $= p + K_{\text{ret}}$, the $p=2$ covariates plus the retained surface tensor columns (near-zero columns dropped): 27=2+25 at $J=4$, 63=2+61 of 64 at $J=7$. The basis *is* over-rich at $n=500$ ($r_{\text{eff}} \approx 21$ of 63 columns, ≈ 8 low-support columns)—**but this does *not* explain the blow-up**: $\text{corr}(\log \widehat{\text{SE}}, -\log \lambda_{\min}) \approx 0$, and the 2 extreme-SE datasets are *not* among the low- $\lambda_{\min}$ (weakest-Gram) ones (coincidence 0/2). We also audited the blow-up at the bootstrap-*resample* level: the extreme- $|\widehat{\beta}^*|$ resamples (top 0.5%) are *indistinguishable* from the bulk in lost-support columns (7.7 vs 7.6), standardized-Gram $\lambda_{\min}$ (1.5 vs 1.6×10^{-3}), and onset diversity (191 vs 195), with all correlations ≈ 0 . So the instability is *not* a Gram-rank or support failure at either the dataset or the resample level; it appears to be a rare numerical/optimization instability of the over-rich small- n sieve rather than a failure of the measured design-rank diagnostics, and we leave the precise trigger *unidentified* (it does not affect the recommended fixed sieve). We therefore present it descriptively (a small- n over-rich-sieve artifact, 2/250 datasets) and *withdraw* any condition-number causal attribution. The recommended fixed sieve showed no analogous instability in the audited runs. (Column big/ $\cap$ rd: the number of datasets with bootstrap $\text{SE} > 3 \times$ the median, and how many of those also fall in the weakest-Gram decile ($\lambda_{\min}$ in the bottom 10%; “weakest-Gram”, not necessarily rank-deficient).)

sieve	n	J	bootstrap SE		degeneracy		$\rho_{\log \text{SE}, -\log \lambda}$	big/ $\cap$ rd
			med	max	r_{eff}	N_{low}		
exact	500	4	0.138	0.20	7/27	2	0.03	0/0
exact	1000	4	0.097	0.13	7/27	2	0.04	0/0
exact	2000	4	0.067	0.09	7/27	2	0.02	0/0
grow	500	7	0.139	96.44	21/63	8	-0.03	2/0
grow	1000	7	0.097	0.12	20/63	7	-0.02	0/0
grow	2000	8	0.069	0.08	26/77	8	0.06	0/0

Table S22: Quantile-crossing on a *dense* 30×30 interior (u, t) grid $\times$ 10 covariate profiles ($\tau \in \{.1, .25, .5, .75, .9\}$, adjacent pairs; $R = 300$). Crossing rates split interior/boundary; “ $\bar{u}/t$ ” is the mean u/t of crossing *incidents* (small $\Rightarrow$ near the low-age boundary). The interior crossing rate is 0 in every cell; boundary crossing rates round to ≈ 0 by $n = 1000$, but at $n=500$ a non-trivial *fraction of replicates* ($\leq 15\%$, Sc. 2) show *some* boundary crossing with per-incident magnitude up to 0.65—so we rearrange by default. cal_{in} is the in-sample τ -calibration error; $\text{cal}_{\text{vis}}^{\text{OOS}}/\text{cal}_{\text{sub}}^{\text{OOS}}$ are the *out-of-sample* errors on an *independent test draw* (same DGP), *visit*-weighted (mean over ≈ 620 test rows) vs. *subject*-equal-weighted (so multi-visit subjects do not dominate). The two agree (0.011–0.022); $\text{cal}_{\text{vis}}^{\text{OOS}}$ is the value *before* rearrangement, and rearrangement shifts it by at most 0.001 in every cell (largest, Sc. 2 $n=500$: $0.0194 \rightarrow 0.0184$; the full before/after pair is in the released **crossing.csv**), so rearrangement does not distort out-of-sample calibration under either weighting.

Sc.	n	cr_{int}	cr_{bnd}	frac. reps	max mag	$\bar{u}/t$	cal_{in}	$\text{cal}_{\text{vis}}^{\text{OOS}}$	$\text{cal}_{\text{sub}}^{\text{OOS}}$
1	500	0.0000	0.0001	0.06	0.178	0.27	0.0017	0.0197	0.0220
1	1000	0.0000	0.0000	0.01	0.057	0.05	0.0011	0.0160	0.0153
1	2000	0.0000	0.0000	0.00	0.000	–	0.0007	0.0127	0.0112
2	500	0.0000	0.0003	0.15	0.652	0.34	0.0018	0.0194	0.0222
2	1000	0.0000	0.0000	0.02	0.201	0.12	0.0011	0.0160	0.0148
2	2000	0.0000	0.0000	0.00	0.000	–	0.0007	0.0122	0.0114

Table S23: Pilot results for the quantile slope $\beta_{\tau,1}$, **IPCW** pipeline (IPCW–A–CV, IPCW–A–Rich, IPCW–B–OS–NCF). Bias is Monte-Carlo over $R = 200$ replicates; coverages are over a 150-replicate bootstrap subset (Monte-Carlo SE ≈ 0.018). “boot” = full-refit subject bootstrap; “froz” = frozen- G_C bootstrap. A separate paired CC-vs-IPCW study is in Table S5 (distinct experiment; the IPCW values here are not reused there).

Sc.	n	τ	$\beta_{\tau,1}$	bias _{ACV}	bias _{BOS}	ESD _{BOS}	cov _{ACV} ^{boot}	cov _{ARich} ^{boot}	cov _{BOS} ^{boot}	cov _{BOS} ^{froz}
1	500	0.10	0.500	0.017	0.018	0.145	0.907	0.927	0.900	0.900
1	500	0.25	0.500	0.001	0.001	0.132	0.880	0.907	0.887	0.887
1	500	0.50	0.500	0.008	0.007	0.106	0.953	0.967	0.960	0.960
1	500	0.75	0.500	-0.023	-0.024	0.111	0.940	0.960	0.947	0.947
1	500	0.90	0.500	-0.005	-0.004	0.139	0.933	0.940	0.953	0.953
1	1000	0.10	0.500	0.007	0.007	0.107	0.900	0.893	0.907	0.907
1	1000	0.25	0.500	0.007	0.007	0.087	0.960	0.947	0.960	0.953
1	1000	0.50	0.500	-0.009	-0.010	0.083	0.947	0.947	0.947	0.947
1	1000	0.75	0.500	0.003	0.004	0.083	0.933	0.953	0.933	0.933
1	1000	0.90	0.500	-0.003	-0.003	0.102	0.927	0.907	0.920	0.920
1	2000	0.10	0.500	-0.002	-0.002	0.071	0.940	0.960	0.947	0.947
1	2000	0.25	0.500	0.005	0.005	0.059	0.960	0.947	0.960	0.960
1	2000	0.50	0.500	-0.000	0.000	0.057	0.927	0.933	0.933	0.933
1	2000	0.75	0.500	0.007	0.007	0.053	0.953	0.987	0.967	0.967
1	2000	0.90	0.500	-0.006	-0.005	0.071	0.953	0.933	0.953	0.953
2	500	0.10	-0.013	0.027	0.028	0.176	0.913	0.900	0.913	0.913
2	500	0.25	0.230	-0.002	-0.003	0.145	0.933	0.947	0.940	0.940
2	500	0.50	0.500	0.005	0.006	0.144	0.953	0.953	0.960	0.960
2	500	0.75	0.770	-0.024	-0.026	0.149	0.947	0.933	0.947	0.947
2	500	0.90	1.013	-0.017	-0.017	0.182	0.933	0.933	0.933	0.927
2	1000	0.10	-0.013	-0.005	-0.005	0.135	0.933	0.920	0.927	0.927
2	1000	0.25	0.230	0.014	0.015	0.097	0.967	0.960	0.973	0.967
2	1000	0.50	0.500	-0.000	0.000	0.110	0.947	0.940	0.947	0.947
2	1000	0.75	0.770	-0.003	-0.003	0.103	0.967	0.967	0.967	0.967
2	1000	0.90	1.013	-0.004	-0.003	0.121	0.953	0.933	0.953	0.953
2	2000	0.10	-0.013	0.000	0.000	0.084	0.960	0.960	0.967	0.967
2	2000	0.25	0.230	-0.006	-0.006	0.073	0.953	0.960	0.953	0.953
2	2000	0.50	0.500	0.005	0.005	0.076	0.927	0.920	0.927	0.927
2	2000	0.75	0.770	-0.007	-0.007	0.075	0.940	0.940	0.933	0.933
2	2000	0.90	1.013	0.009	0.009	0.091	0.960	0.947	0.947	0.947

Table S24: Non-proportional-hazards (log-normal AFT) onset, **IPCW–A–CV** pipeline (as in Table S23). Bias and bootstrap coverage of $\beta_{\tau,1}$ are broadly similar to the Cox-Weibull results of Table S23, supporting robustness to the onset mechanism (the method never models onset). $R = 200$ for bias; coverage over a 60-replicate subset only (Monte-Carlo SE ≈ 0.028), so this is a *pilot-grade* stress test — the several 0.90–0.917 cells are not distinguishable from nominal at this R . The -0.050 bias at (Sc. 2, $n=500$, $\tau=0.9$) has ESD ≈ 0.18 (bias MCSE ≈ 0.023), so it is ≈ 2 MCSE and not yet alarming; a CC–A–CV AFT run and a larger R are deferred to the consolidated specification.

Sc.	n	τ	$\beta_{\tau,1}$	bias _{ACV}	cov _{ACV} ^{boot}
1	500	0.100	0.500	0.003	0.950
1	500	0.500	0.500	-0.003	0.933
1	500	0.900	0.500	-0.007	0.933
1	1000	0.100	0.500	-0.005	0.917
1	1000	0.500	0.500	0.008	0.917
1	1000	0.900	0.500	-0.009	0.983
1	2000	0.100	0.500	0.002	0.950
1	2000	0.500	0.500	-0.007	0.983
1	2000	0.900	0.500	-0.008	0.967
2	500	0.100	-0.013	0.008	0.967
2	500	0.500	0.500	0.008	0.933
2	500	0.900	1.013	-0.050	0.900
2	1000	0.100	-0.013	0.005	0.967
2	1000	0.500	0.500	-0.016	0.983
2	1000	0.900	1.013	-0.018	0.950
2	2000	0.100	-0.013	0.011	0.900
2	2000	0.500	0.500	0.005	0.950
2	2000	0.900	1.013	-0.005	0.900

Table S25: Computational reproducibility audit, a *dedicated reduced-B run* ($R = 300$ per cell, inner $B=60$; *not* a summary of the headline $B=500$ log), $\tau \in \{.25, .5, .75\}$. Failure/fallback rates that would bias the headline numbers if silently dropped, for *both* the CC and IPCW pipelines: the QR (QR-solver) and CF (B-OS-CF) failure columns are shown *separately for both pipelines*; the remaining three columns— $\hat{f}$ -cap, $\kappa(\hat{S})$, and the inner subject-bootstrap replicate-failure rate (“boot”)—are *CC-pipeline* diagnostics (the recommended estimator). *Failed replicates are counted, not discarded*. We also report the bread condition-number distribution (median, 95th pct) rather than only a permissive (and loose) $\kappa > 10^8$ binary flag. *Denominators*: QR and CF are over the $R=300$ datasets; $\hat{f}$ -cap over all fold-visit density estimates; boot over the $R \times B$ inner resamples. **The QR and B-OS-CF structural-fit failure rates are 0 for both the CC and IPCW pipelines**, and the (CC) bootstrap-replicate failure rate is 0 (we tabulate the CC bootstrap-failure column only, so we do *not* claim an IPCW bootstrap-failure audit here); a zero-event rate at $R=300$ has a 95% binomial upper bound of $\approx 1\%$, not an exact 0. $\kappa(\hat{S})$ stays small (median ≈ 4 , 95th pct ≤ 9.1), so ill-conditioning is not a concern; the only non-zero mode is the $\hat{f}$ -cap ($\leq 1.0\%$). *Scope*: base CC/IPCW pipelines at the central quartile quantiles only; the degenerate small- n growing-sieve bootstrap (Table S12) is *not* covered here and needs a separate distributional audit.

Sc.	n	τ	CC		IPCW		$\kappa(\hat{S})$		$\hat{f}$ -cap	boot
			QR	CF	QR	CF	med	q_{95}		
1	500	0.25	0.000	0.000	0.000	0.000	4.0	6.5	0.009	0.000
1	500	0.50	0.000	0.000	0.000	0.000	4.1	6.1	0.009	0.000
1	500	0.75	0.000	0.000	0.000	0.000	4.1	6.5	0.008	0.000
1	1000	0.25	0.000	0.000	0.000	0.000	4.0	5.2	0.003	0.000
1	1000	0.50	0.000	0.000	0.000	0.000	3.9	5.0	0.003	0.000
1	1000	0.75	0.000	0.000	0.000	0.000	3.9	5.1	0.003	0.000
1	2000	0.25	0.000	0.000	0.000	0.000	3.9	4.6	0.001	0.000
1	2000	0.50	0.000	0.000	0.000	0.000	3.8	4.4	0.001	0.000
1	2000	0.75	0.000	0.000	0.000	0.000	3.8	4.5	0.001	0.000
2	500	0.25	0.000	0.000	0.000	0.000	4.9	9.0	0.010	0.000
2	500	0.50	0.000	0.000	0.000	0.000	4.9	8.6	0.009	0.000
2	500	0.75	0.000	0.000	0.000	0.000	4.9	9.1	0.010	0.000
2	1000	0.25	0.000	0.000	0.000	0.000	4.4	6.5	0.003	0.000
2	1000	0.50	0.000	0.000	0.000	0.000	4.3	6.2	0.003	0.000
2	1000	0.75	0.000	0.000	0.000	0.000	4.4	6.9	0.004	0.000
2	2000	0.25	0.000	0.000	0.000	0.000	4.2	5.1	0.001	0.000
2	2000	0.50	0.000	0.000	0.000	0.000	4.2	5.2	0.001	0.000
2	2000	0.75	0.000	0.000	0.000	0.000	4.2	4.9	0.001	0.000

Table S26: **Distributional effect recovery and scalar-slope variability (Sun’s real estimator vs. ours)**. Same Monte-Carlo datasets under Sun et al.’s survival/truncation/fixed-visit design ($R = 500$; common interior domain). Sun’s profile partial-likelihood slope $\hat{\beta}_1$ (their compiled `marker3.cpp`) and its Monte-Carlo ESD vs. the ESD of our median slope $\hat{\beta}_{.5,1}$, plus our $\beta_{\tau,1}$ bias at $\tau = 0.1, 0.5, 0.9$ and the recovered tail spread $\hat{\beta}_{.9} - \hat{\beta}_{.1}$. *Both estimators are unbiased when their target is well-defined*. In DGP1 the direct median slope has smaller Monte-Carlo variability than the feasible Sun slope for the shared scalar target ($\sim 4\times$ smaller ESD); this is a point-estimation comparison, not an inferential-efficiency claim. An oracle- α decomposition shows Sun’s feasible $\hat{\beta} = \hat{\theta} - \hat{\alpha}$ (Thm. 3.1) is *not* less stable because it estimates α : the oracle- α version $\hat{\theta} - \alpha_0$ is in fact *more* variable, so the dominant variability appears to come from the profile-kernel marker score $\hat{\theta}$, with $\hat{\theta} - \hat{\alpha}$ partly benefiting from covariance cancellation. *The Sun-ESD vs. Q-ESD comparison is like-for-like only in DGP1, where the mean and median slopes coincide ($\beta_{\text{mean}} = \beta_{.5}$); in DGP2–DGP4 the scalar targets differ, and in DGP5 the conditional mean is undefined, so Sun’s slope variability is reported only as a fragility diagnostic, not an efficiency comparison*. DGP2 only the quantile slope sees the covariate-driven dispersion; DGP3 no proportional slope exists; DGP4 the effect is confined to the upper tail ($\beta_{.9} \gg \beta_{.5}$) so Sun’s scalar hides it; in DGP5 (heavy t_5 log-tail, mean ill-posed) Sun’s raw ESD is unstable and non-monotone, driven by rare explosive observations, while robust spread summaries shrink with n (Table S29) and our median slope is robust.

n	Sun $\hat{\beta}_1$ (ESD)	Q ESD _{.5}	Q $\beta_{\tau,1}$ bias			spread $\beta_{.9}-\beta_{.1}$
			.10	.50	.90	Q / true
DGP1 proportional, homoscedastic mean						
500	0.51 (0.10)	0.024	-0.002	-0.001	0.002	0.00/0.00
1000	0.50 (0.07)	0.017	0.001	0.000	0.001	0.00/0.00
2000	0.50 (0.05)	0.012	0.001	0.001	0.001	0.00/0.00
DGP2 proportional mean, dispersion varies						
500	0.50 (0.15)	0.088	0.008	-0.003	-0.019	1.13/1.15
1000	0.50 (0.11)	0.063	0.010	0.001	-0.001	1.14/1.15
2000	0.50 (0.08)	0.047	0.000	-0.006	-0.006	1.15/1.15
DGP3 non-proportional mean (headline)						
500	0.77 (0.14)	0.063	0.013	-0.001	-0.014	1.13/1.15
1000	0.76 (0.10)	0.046	0.008	0.001	-0.006	1.14/1.15
2000	0.77 (0.07)	0.032	0.007	0.001	-0.002	1.14/1.15
DGP4 tail-specific effect						
500	0.67 (0.20)	0.064	0.004	0.005	-0.111	0.98/1.10
1000	0.67 (0.14)	0.047	-0.002	0.003	-0.051	1.05/1.10
2000	0.67 (0.09)	0.033	0.000	-0.001	-0.031	1.07/1.10
DGP5 heavy-tailed error (mean undefined)						
500	0.53 (0.49)	0.054	0.000	0.001	0.002	0.00/0.00
1000	0.50 (0.09)	0.037	0.002	0.002	0.002	0.00/0.00
2000	0.52 (0.37)	0.028	-0.002	-0.001	0.000	0.00/0.00

Table S27: **Conditional-mean contrast — the headline** ($\text{IMSE} \times 10^3$). Both methods target the conditional mean. Sun: the *basic* proportional model ($E[Y|u, t, x] = \mu_0(u, t)e^{\beta'x}$, log-contrast $= \hat{\beta}_1$, constant) and his Sec. 3.5 *stratified-baseline* extension (Sun-S: his genuine estimator fit separately per X_1 group, giving a nonparametric (u, t) -varying contrast). Ours: the model-free quantile integral Q-INT ($\int_0^1 e^{Q_\tau} d\tau$) and the tail-stable retransformation Q-LN ($\hat{Q}_{.5} + \frac{1}{2}\hat{\sigma}^2$). IMSE of the mean log-contrast $\Delta(u, t)$. Under non-proportionality (DGP3) Δ truly varies: Sun-basic collapses it to a pseudo-average, and the fully stratified Sun-S recovers it only at a large *variance* cost (two nonparametric surfaces), so its IMSE exceeds Sun-basic and far exceeds Q (both Q reconstructions several-fold lower; the paired Q-LN – Sun-basic difference is significantly negative, MCSE in parentheses). Sun-S is a fully stratified implementation and is *not* intended to represent all prespecified low-dimensional interaction bases. In DGP1 the lower Q IMSE reflects the smaller Monte-Carlo variability of the direct shared scalar slope (Table S26); in DGP2 Q-LN benefits from the log-location-scale structure, whereas the model-free Q-INT is more tail-sensitive and is *not* uniformly better than Sun (Q-INT – Sun $= +2.5 \times 10^{-3}$ at $n=2000$). In **DGP4 Sun-basic is *best*** (well matched to the proportional mean target). DGP5’s mean is undefined (–).

n	contrast IMSE $\times 10^3$				paired –Sun $\times 10^3$ (MCSE)	
	Sun	Sun-S	Q-LN	Q-INT	Q-LN	Q-INT
DGP1 proportional, homoscedastic mean						
500	9.445	93.959	0.360	0.353	-9.084 (0.624)	-9.092 (0.624)
1000	5.084	42.660	0.183	0.184	-4.901 (0.312)	-4.900 (0.312)
2000	2.329	20.742	0.091	0.091	-2.238 (0.151)	-2.239 (0.151)
DGP2 proportional mean, dispersion varies						
500	22.763	188.000	8.334	14.944	-14.429 (1.434)	-7.820 (1.558)
1000	12.612	102.526	4.546	9.597	-8.066 (0.710)	-3.015 (0.797)
2000	5.853	59.003	2.306	8.374	-3.547 (0.350)	2.522 (0.452)
DGP3 non-proportional mean (headline)						
500	23.703	158.101	4.398	7.782	-19.305 (1.275)	-15.921 (1.338)
1000	14.290	78.095	2.169	5.104	-12.121 (0.580)	-9.187 (0.634)
2000	9.539	41.998	1.043	3.876	-8.496 (0.320)	-5.663 (0.373)
DGP4 tail-specific effect						
500	40.235	239.699	53.970	27.912	13.734 (3.103)	-12.323 (2.463)
1000	18.489	139.674	40.737	14.536	22.247 (1.794)	-3.953 (1.210)
2000	8.761	72.335	36.858	9.927	28.097 (1.239)	1.166 (0.695)
DGP5 heavy-tailed error (mean undefined)						
500	–	–	–	–	–(–)	–(–)
1000	–	–	–	–	–(–)	–(–)
2000	–	–	–	–	–(–)	–(–)

Table S28: **Conditional-mean surface and Sun bandwidth sensitivity** ($\log\text{-IMSE} \times 10^3$, $x_1=0$). Sun’s kernel baseline $\hat{\mu}_0$ at his density-reference bandwidth h ($2.34\text{sd}(A)N^{-1/6}$) and at the regression bandwidth $0.5h$, vs. our Q-LN and Q-INT reconstructions. Sun’s surface is bandwidth-sensitive—the density rule over-smooths the steep baseline (h column), and $0.5h$ roughly halves the IMSE—so the Sun-vs-Q surface gap is partly a smoothing-bandwidth effect, not a method deficiency; the bandwidth-free slope and contrast (Tables S26–S27) are the substantive comparisons. DGP5’s mean is undefined (–).

n	R	Sun $\hat{\mu}_0$		ours	
		h	$0.5h$	Q-LN	Q-INT
DGP1 proportional, homoscedastic mean					
500	500	31.021	15.380	1.081	0.967
1000	500	18.505	8.316	0.695	0.607
2000	500	10.757	4.509	0.519	0.442
DGP2 proportional mean, dispersion varies					
500	500	36.480	29.638	5.053	5.314
1000	500	22.935	18.114	2.625	2.948
2000	500	12.885	10.055	1.428	1.826
DGP3 non-proportional mean (headline)					
500	500	40.442	30.410	2.400	2.411
1000	500	24.352	17.015	1.217	1.293
2000	500	15.570	10.565	0.754	0.848
DGP4 tail-specific effect					
500	500	43.106	39.571	4.737	4.582
1000	500	26.898	24.246	2.388	2.566
2000	500	15.794	13.366	1.305	1.482
DGP5 heavy-tailed error (mean undefined)					
500	500	—	—	—	—
1000	500	—	—	—	—
2000	500	—	—	—	—

Table S29: **DGP5 heavy-tail robustness diagnostics** (Student- t_5 log-error; true location shift $\beta_1 = 0.5$). Sun’s mean-slope sampling distribution over $R = 500$ replicates vs. our median slope. The ordinary ESD is large and non-monotone in n , but the median, IQR, 95th percentile, and 3-IQR-trimmed ESD (tESD) are small and *shrink* with n : the inflated ESD is driven by a few replicates with an extreme maximum marker (the t_5 log-tail produces markers exceeding 10^4 in rare replicates), not a coding artifact. Our median slope is unbiased with a small, shrinking ESD throughout. This is an estimand-validity illustration: the conditional mean (Sun’s target) is undefined under a t_5 log-tail, so it is not a fair mean-estimation contest.

n	Sun $\hat{\beta}_1$		Sun $\hat{\beta}_1$ robust spread				
	mean	ESD	med	IQR	$q_{.95}$	tESD	Q ESD _{.5}
500	0.53	0.49	0.51	0.15	0.69	0.12	0.054
1000	0.50	0.09	0.50	0.11	0.63	0.08	0.037
2000	0.52	0.37	0.50	0.08	0.60	0.06	0.028